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PAPER

A unified model for species fluxes, ion impact energy and heat transfer at the electrode of an atmospheric-pressure arc discharge

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Abstract

1 Introduction

Arc discharges are widely used in welding, cutting and thermal spraying, and in recent years they have found new applications in plasma pyrolysis, advanced material synthesis and metallurgical reduction [1, 2, 3, 4]. An atmospheric-pressure arc spans length scales that differ by several orders of magnitude. Most of the discharge is the hot, quasi-neutral arc column, millimeters to centimeters in size, in which convection, thermal conduction, Joule heating and radiation interact under conditions often close to local thermodynamic equilibrium (LTE). Within a fraction of a millimeter of the electrodes, by contrast, the plasma is far from equilibrium. There it is in contact with relatively cold, absorbing electrode surfaces, which produce steep gradients in the species densities and temperatures. In sub-micrometer layers next to the surfaces, charge separation develops, quasi-neutrality breaks down and strong electric fields arise.

Resolving this full range of length scales over the entire discharge is neither computationally practical nor necessary for most applications. Poisson's equation, which governs the charge separation and the electric field near the electrodes, need not be solved in the bulk plasma, where quasi-neutrality holds. Conversely, the near-electrode layers are thin compared with the flow and geometric scales of the arc, so convection, electrode jets and the electrode geometry can be neglected there; these layers are dominated by diffusion, reactions and sheath effects. Arc models therefore typically combine a quasi-neutral model of the bulk plasma with a separate near-electrode model. Several strategies have been adopted for the bulk plasma, and the choice determines what the near-electrode model must describe.

(i) Local thermodynamic equilibrium. Measurements of the species temperatures and the electron density indicate that the arc core is generally close to LTE [5, 6, 7]. In an LTE model, electrons and heavy particles share a single temperature T and have Maxwellian velocity distributions; Boltzmann excitation, Saha ionization and Dalton's law then determine the plasma state and its properties. The governing equations typically consist of mass, momentum and energy conservation together with current continuity [8, 9, 10]. These models are computationally efficient because the species densities and transport properties need not be solved for: they follow from the local temperature and pressure. An LTE bulk model, however, requires a near-electrode model that captures the separation of the electron and heavy-particle temperatures, the departure from Saha equilibrium, the breakdown of quasi-neutrality and the formation of the space-charge sheath [11, 12, 13].

(ii) Two-temperature models. Two-temperature (2T) models treat electrons and heavy particles as separate Maxwellian populations at temperatures T_e and T_h, coupled through elastic energy exchange. In two-temperature equilibrium (2T-EQ) models, the composition is in addition obtained from a generalized Saha relation for $T_e \neq T_h$. Hsu and Pfender [14] showed that $T_e \approx T_h$ in the high-density arc core, with significant temperature separation in the low-density fringes and near the anode. The generalized Saha closure is not unique, however: different formulations can predict electron densities that differ by up to an order of magnitude once T_e/T_h departs from unity [15]. Freton *et al.* [16] have further shown that several published formulations distribute the ionization term and the reactive thermal conductivity inconsistently between the electron and

heavy-particle energy equations. A 2T bulk model still requires a near-electrode model that captures the ionization layer, or presheath, where Saha equilibrium is violated, as well as the breakdown of quasi-neutrality and the formation of the space-charge sheath [17, 18].

(iii) Nonequilibrium models. Fully nonequilibrium models abandon equilibrium closures such as LTE and generalized Saha equilibrium, and let the plasma composition and the electron and heavy-particle temperatures evolve independently. They typically include multicomponent species transport and separate electron and heavy-particle energy equations. Examples are the nonequilibrium arc models of Haidar [19], Gonzalez *et al.* [20] and Amakawa *et al.* [21]. Because they resolve departures from equilibrium directly, these models can describe the transition from the LTE-like bulk plasma to the strongly nonequilibrium electrode region. The space-charge sheath, however, is still treated separately, by a sheath model matched to the quasi-neutral plasma [22].

A widely used near-electrode approach treats the ionization layer, or presheath, as a collisional region and couples it to a sheath that is assumed to be collisionless and is described analytically [23, 24, 18, 17]. This avoids solving the stiff Poisson equation across the sheath. Almeida, Benilov and Naidis [11] estimated the number of collisions that an ion undergoes while crossing the cathode space-charge sheath of an atmospheric-pressure argon arc, and found that this number is comparable to unity rather than small. Strictly speaking, the collisionless-sheath assumption is therefore unjustified. A kinetic (particle-in-cell/Monte Carlo collision) study of the near-cathode region by Sun *et al.* [25] likewise supports the view that this region, sheath included, is collision-dominated. At the cathode the consequences are limited: the integral characteristics of the plasma–cathode interaction predicted by unified and sheath-matched models are generally in good agreement [11, 26]. At the anode, however, Semenov *et al.* [13] showed that the common collisionless sheath model, in which the ion velocity at the sheath edge is set to the Bohm velocity, fails. A physically consistent electrode model for high-pressure arcs should therefore account for collisional ion transport within the sheath.

The alternative is *unified modeling*, in which a single set of governing equations, including Poisson's equation, is solved across the whole near-electrode region [11, 13, 25, 27]. Ion collisions in the sheath are then accounted for by construction. Because a unified model captures both thermal and ionization nonequilibrium, it can be matched directly to an LTE model of the bulk plasma. Nor do field reversals need to be anticipated. Field reversals and changes in the sign of the potential drop near the anode have been reported in atmospheric-pressure arcs, and they depend on several factors, including the attachment mode, the current density and electrode ablation [28, 29, 30, 22]. When the drift-diffusion–Poisson system is solved self-consistently across the near-electrode region, the potential structure of the sheath and its polarity emerge from the solution instead of being assumed a priori. Unified models are computationally expensive, however, because the mesh must resolve the very small length scales of the sheath. Consequently, their application to arc discharges has largely been restricted to one-dimensional calculations extending from the electrode into the bulk plasma [11, 12, 31, 13].

In this paper we adopt the decomposition described above, treating the quasi-neutral bulk plasma and the near-electrode region with separate models, and present a transient near-electrode model intended to supply the electrode boundary conditions for our LTE model of the bulk arc [32]. The near-electrode model is a unified drift-diffusion–Poisson formulation that determines simultaneously the charged-particle distributions, the species fluxes, the sheath voltage drop and the heat flux to the electrode. The same system of equations is solved continuously from the quasi-neutral bulk plasma at one end of the domain, through the near-electrode transition region, and into the sheath. In previous unified models [11, 31, 12, 13], the bulk-plasma state at the outer boundary was taken from the asymptotic arc column, in which Joule heating is balanced by radiation alone, without radial losses. The computational domain therefore had to extend far into the arc, where the temperature and density gradients could be assumed to vanish. In the present model, the bulk state is instead supplied at the outer boundary, from measurements or from an LTE model of the arc, and radial losses are included so that this state is consistent with the energy balance of the column. Gradients of the plasma properties may exist at this boundary; the only requirement is that the plasma there be reasonably close to LTE. The domain therefore need not extend far into the arc, which is especially helpful in multidimensional simulations. The disparate length scales involved (Section 2.1) are resolved with adaptive mesh refinement (AMR), with the mesh in the sheath refined to a fraction of the local Debye length. The model and its numerical implementation are described in Section 2. The model is validated in Section 3 and applied to the near-cathode and near-anode layers of a high-intensity argon arc in Section 4.

Table 1. Characteristic length scales of LTE argon at atmospheric pressure ($T_e = T_h = T$, Saha equilibrium), ordered from left to right by increasing size; $\lambda_i \approx \lambda_e$, and λ_{rxn} and λ_T are comparable from 11 000 to 14 000 K. The reaction scales use the rate coefficients of Almeida et al. [11].

T (K)	λ_{De} (m)	λ_i (m)	λ_e (m)	λ_{T_h} (m)	λ_{rxn} (m)	λ_T (m)
11 000	3.9×10^{-8}	9.5×10^{-7}	1.8×10^{-6}	1.8×10^{-6}	2.6×10^{-4}	3.4×10^{-4}
12 000	3.0×10^{-8}	8.1×10^{-7}	1.2×10^{-6}	2.1×10^{-6}	1.8×10^{-4}	2.3×10^{-4}
14 000	2.1×10^{-8}	6.6×10^{-7}	7.5×10^{-7}	3.2×10^{-6}	1.1×10^{-4}	1.4×10^{-4}
18 000	2.1×10^{-8}	8.8×10^{-7}	8.9×10^{-7}	5.9×10^{-6}	6.6×10^{-5}	1.7×10^{-4}
20 000	2.3×10^{-8}	1.1×10^{-6}	1.1×10^{-6}	6.9×10^{-6}	5.1×10^{-5}	2.2×10^{-4}
22 000	2.5×10^{-8}	1.4×10^{-6}	1.4×10^{-6}	7.8×10^{-6}	4.0×10^{-5}	2.8×10^{-4}
23 000	2.6×10^{-8}	1.6×10^{-6}	1.6×10^{-6}	8.2×10^{-6}	3.7×10^{-5}	3.1×10^{-4}

2 Physical model and scale analysis

2.1 Spatial scale analysis

In the electrode region, the plasma is in contact with a relatively cold, absorbing electrode surface. The resulting strong gradients drive the plasma progressively further from LTE as the electrode is approached. We estimate the length scales over which these departures develop in a high-intensity atmospheric-pressure arc by considering the LTE plasma adjacent to the electrode region, at temperatures from 23 000 K, typical of the cathode region, down to 11 000 K, typical of the anode region [10, 33].

In the bulk plasma, the electron number density n_e is governed by ionization and recombination. Let n_n be the number density of argon atoms, $n_i = n_e$ that of the ions, and k_{ion} and k_{rec} the rate coefficients of electron-impact ionization and of three-body recombination, for which we use those of Almeida et al. [11] (equation (2)). The electron density then evolves as $\dot{n}_e = k_{\mathrm{ion}} n_e n_n - k_{\mathrm{rec}} n_e^3$. Linearizing this rate about equilibrium at fixed pressure p and temperature T, for which Dalton's law gives $\delta n_n = -2\,\delta n_e$, yields the reaction time $\tau_{\mathrm{rxn}} = 1/(2k_{\mathrm{rec}} n_e^2 + 2k_{\mathrm{ion}} n_e)$. Over this time, the charged particles diffuse ambipolarly over the reaction length $\lambda_{\mathrm{rxn}} = \sqrt{D_a \tau_{\mathrm{rxn}}}$, where D_a is the ambipolar diffusion coefficient (Appendix A). A plasma displaced from ionization equilibrium therefore returns to it over a distance of order λ_{rxn}.

The electron and heavy-particle temperatures, in turn, equilibrate through elastic collisions. In an elastic collision with a heavy particle, an electron loses only a fraction $2m_e/M$ of its energy, where m_e and M are the electron and argon masses. Its energy therefore relaxes only after about $M/(2m_e)$ collisions, over which it random-walks a distance $\lambda_T = \lambda_e/\sqrt{2m_e/M}$; this is the length scale of energy relaxation between the electrons and the heavy particles. Here $\lambda_e = 1/(n_n \overline{Q}_{en} + n_i \overline{Q}_{ei})$ is the electron mean free path, and $\overline{Q}_{en}$ and $\overline{Q}_{ei}$ are the averaged momentum-transfer cross sections for electron–atom and electron–ion collisions (Appendix A): $\overline{Q}_{en}$ is the Maxwellian average of the argon momentum-transfer cross section at T_e, and $\overline{Q}_{ei}$ is the Coulomb cross section. Ions and atoms, by contrast, have nearly equal masses and exchange energy efficiently, so their temperatures equalize over $\lambda_{T_h} = 1/[(n_i + n_n)\overline{Q}_{in}]$, where $\overline{Q}_{in}$ is the ion–atom momentum-transfer cross section, dominated by resonant charge exchange (equation (41)).

At the smallest scale, the electron Debye length $\lambda_{De} = \sqrt{\varepsilon_0 k_B T_e/(n_e e^2)}$ is of order 10^{-8} m; it sets the extent of charge separation and space-charge effects. Here ε_0 is the vacuum permittivity, k_B the Boltzmann constant and e the elementary charge. These scales are listed in table 1, together with the electron mean free path λ_e and the ion mean free path $\lambda_i = 1/(n_n \overline{Q}_{in} + n_i \overline{Q}_{ii})$, where $\overline{Q}_{ii}$ is the Coulomb cross section $\overline{Q}_{ei}$ evaluated at T_h.

The resulting hierarchy of length scales is $\lambda_{De} \ll \lambda_i \approx \lambda_e \lesssim \lambda_{T_h} \ll \lambda_{\mathrm{rxn}} \approx \lambda_T$. The largest scales, λ_{rxn} and λ_T, indicate that the electrode affects the plasma over a region of thickness of order 10^{-4} m. Other models and analyses [11, 23, 17, 24, 34, 13] arrive at similar estimates. Consistently with these estimates, Lowke and Tanaka [35] found that departures from LTE are confined to within about 0.1 mm of the electrodes of a 200 A atmospheric-pressure argon arc, and suggested that the rest of the plasma can be modeled as being in LTE. Because λ_e and λ_i are small compared with this region, a continuum description with Maxwellian velocity distributions is reasonable over most of it; because λ_{T_h} is small as well, ions and atoms can be described by a single temperature T_h. The validity of both assumptions close to the electrodes is examined a posteriori in Section 4.3.

2.2 Governing equations

Externally imposed bulk flows, such as the cathode jet, are not modeled in the electrode region. The heavy gas is nevertheless not assumed to be at rest, since it can move during the transient stages of the discharge. The plasma is taken to consist of three species: electrons (e), singly charged argon ions (i), and argon atoms (n). Multiply charged ions are neglected: in the anode

region the temperature is too low to produce them, and in the cathode region they are present but have a negligible effect [36]. Each species is assumed to follow a Maxwellian velocity distribution (Appendix B). The unknowns are the species number densities, the electron and heavy-particle temperatures, the heavy-gas velocity and the electric field, which are governed by the following equations.

The number densities of the three species obey

$$\frac{\partial n_e}{\partial t} + \nabla \cdot \mathbf{\Gamma}_e = S, \qquad \frac{\partial n_i}{\partial t} + \nabla \cdot \mathbf{\Gamma}_i = S, \qquad \frac{\partial n_n}{\partial t} + \nabla \cdot \mathbf{\Gamma}_n = -S, \tag{1}$$

where $\mathbf{\Gamma}_s = n_s \mathbf{u}_s$ is the flux of species s and S is the net ionization rate. The latter accounts for electron-impact ionization and its inverse, three-body recombination with an electron as the third particle,

$$S = k_{\text{ion}}\, n_e n_n - k_{\text{rec}}\, n_e^2 n_i. \tag{2}$$

Different authors have evaluated the rate coefficients k_{ion} and k_{rec} in different ways [11, 23, 35, 37]. In this work we use those of Almeida et al. [11], taken from Benilov and Naidis [38]. In this model, ionization proceeds mainly stepwise, through the first excited level; the escape of resonance radiation, which reduces the ionization rate, is accounted for when $n_e \lesssim 10^{21}$ m^{-3}; and k_{rec} follows from k_{ion} by detailed balance with Saha equilibrium at T_e.

The momentum equation for the plasma as a whole is $\rho\, D\mathbf{u}/Dt = \mathbf{F}$, where $\mathbf{F}$ is the net force on the plasma. For each species s, the corresponding equation is $m_s n_s\, D_s\mathbf{u}_s/Dt = \mathbf{F}_s$, where $\mathbf{F}_s$ is the net force on that species. We neglect the inertia of each species relative to the heavy gas, $D_s\mathbf{u}_s/Dt \simeq D\mathbf{u}/Dt$, so that every species shares the acceleration of the heavy gas and there is no relative acceleration between species. The inertia of species s is then $Y_s\mathbf{F}$, where $Y_s = m_s n_s/\rho$ is its mass fraction, and the species momentum equations reduce to the local force balances [39]

$$0 = -en_e\mathbf{E} - \nabla p_e - m_e n_e \bar{\nu}_{en}^M(\mathbf{u}_e - \mathbf{u}_n) - m_e n_e \bar{\nu}_{ei}^M(\mathbf{u}_e - \mathbf{u}_i) - C_e n_e k_B \nabla T_e - Y_e\mathbf{F}, \tag{3}$$

$$0 = +en_i\mathbf{E} - \nabla p_i - Mn_i \bar{\nu}_{in}^M(\mathbf{u}_i - \mathbf{u}_n) - Mn_i \bar{\nu}_{ie}^M(\mathbf{u}_i - \mathbf{u}_e) + C_i n_e k_B \nabla T_e - Y_i\mathbf{F}, \tag{4}$$

$$0 = -\nabla p_n - Mn_n \bar{\nu}_{ni}^M(\mathbf{u}_n - \mathbf{u}_i) - Mn_n \bar{\nu}_{ne}^M(\mathbf{u}_n - \mathbf{u}_e) + C_n n_e k_B \nabla T_e - Y_n\mathbf{F}. \tag{5}$$

The frequencies $\bar{\nu}_{en}^M$, $\bar{\nu}_{ei}^M$, $\bar{\nu}_{in}^M$, $\bar{\nu}_{ie}^M$, $\bar{\nu}_{ni}^M$, and $\bar{\nu}_{ne}^M$ are *averaged momentum-transfer collision frequencies*, C_e is the electron thermal-force coefficient, and C_i and C_n are its reactions on the ions and atoms; their calculation is described in Appendix B. Momentum conservation in each collision pair relates the frequencies, and the thermal forces sum to zero:

$$m_e n_e \bar{\nu}_{en}^M = Mn_n \bar{\nu}_{ne}^M, \qquad m_e n_e \bar{\nu}_{ei}^M = Mn_i \bar{\nu}_{ie}^M, \qquad n_i \bar{\nu}_{in}^M = n_n \bar{\nu}_{ni}^M, \qquad C_i + C_n = C_e. \tag{6}$$

Adding equations (3)–(5) and using equation (6) gives the total momentum equation,

$$\rho\, D\mathbf{u}/Dt = e(n_i - n_e)\mathbf{E} - \nabla p. \tag{7}$$

In practice, equations (3), (4) and (7) are solved instead of (5), and the neutral flux is obtained from the definition of the heavy-gas velocity $\mathbf{u}$ as the mass-weighted average of the ion and neutral velocities,

$$\mathbf{\Gamma}_n = (n_i + n_n)\mathbf{u} - n_i\mathbf{u}_i. \tag{8}$$

Solving the electron balance (3) for the electron flux, with $\bar{\nu}_e^M = \bar{\nu}_{en}^M + \bar{\nu}_{ei}^M$ the total electron momentum-transfer frequency and $\nabla p_e = k_B T_e \nabla n_e + n_e k_B \nabla T_e$, gives

$$\mathbf{\Gamma}_e = -\frac{en_e}{m_e \bar{\nu}_e^M}\mathbf{E} - \frac{k_B T_e}{m_e \bar{\nu}_e^M}\nabla n_e - (1 + C_e)\,\frac{n_e k_B}{m_e \bar{\nu}_e^M}\nabla T_e + \frac{n_e}{\bar{\nu}_e^M}\left(\bar{\nu}_{en}^M \mathbf{u}_n + \bar{\nu}_{ei}^M \mathbf{u}_i\right), \tag{9}$$

where the negligible term $Y_e\mathbf{F}$ has been dropped. The first two terms are the conventional drift and diffusion. Their coefficients, $e/(m_e\bar{\nu}_e^M)$ and $k_B T_e/(m_e\bar{\nu}_e^M)$, are the magnitudes of the electron mobility and diffusivity; because both are set by the same frequency, they satisfy the Einstein relation identically. The third term combines the temperature-gradient part of the electron pressure force with the thermal force. The last term is the collisional drag exerted on the electrons by the moving heavy species.

The ion flux follows from the ion balance (4) in the same way. With $\bar{\nu}_i^M = \bar{\nu}_{in}^M + \bar{\nu}_{ie}^M$ the total ion momentum-transfer frequency and $\nabla p_i = k_B T_h \nabla n_i + n_i k_B \nabla T_h$,

$$\mathbf{\Gamma}_i = \frac{en_i}{M\bar{\nu}_i^M}\mathbf{E} - \frac{k_B T_h}{M\bar{\nu}_i^M}\nabla n_i - \frac{n_i k_B}{M\bar{\nu}_i^M}\nabla T_h + C_i\,\frac{n_e k_B}{M\bar{\nu}_i^M}\nabla T_e - \frac{n_i}{n_h}\frac{\mathbf{F}}{M\bar{\nu}_i^M} + \frac{n_i}{\bar{\nu}_i^M}\left(\bar{\nu}_{in}^M \mathbf{u}_n + \bar{\nu}_{ie}^M \mathbf{u}_e\right). \tag{10}$$

The first three terms are ion drift, diffusion, and diffusion driven by the heavy-particle temperature gradient, with mobility $e/(M\bar{\nu}_i^M)$ and diffusivity $k_B T_h/(M\bar{\nu}_i^M)$. The fourth term is the reaction of the ions to the electron thermal force. The fifth term is the ions' share, $Y_i\mathbf{F}$, of the net force on the plasma, which accelerates the ions together with the rest of the heavy gas. The last term is the collisional drag exerted on the ions by the atoms and electrons.

The velocities in the drag terms of equations (9) and (10) are those of the other species, so the two fluxes are coupled. If the drag, thermal-force, and mixture-acceleration terms are removed ($C_e = C_i = 0$, $Y_i\mathbf{F} = 0$, and $\mathbf{u}_s = \mathbf{u} = 0$ in the drag terms), equations (9) and (10) reduce to the classical drift–diffusion fluxes, with mobilities $e/(m_e\bar{\nu}_e^M)$ and $e/(M\bar{\nu}_i^M)$. Unlike the classical fluxes, however, they retain the full pressure-gradient force of each species in a non-isothermal plasma.

The electric field is obtained from Poisson's equation,

$$\nabla\cdot(\varepsilon_0\mathbf{E}) = e(n_i - n_e), \qquad \mathbf{E} = -\nabla\phi, \tag{11}$$

which is solved over the entire computational domain. The transition from the quasi-neutral bulk plasma to the space-charge region is thereby captured without imposing the sheath boundaries a priori.

The electron thermal energy density $\mathcal{E} = \frac{3}{2}n_e k_B T_e$ obeys

$$\frac{\partial\mathcal{E}}{\partial t} + \nabla\cdot\mathbf{h}_e = -e\,\mathbf{\Gamma}_e\cdot\mathbf{E} - \frac{3}{2}n_e k_B(T_e - T_h)\,\frac{2m_e}{M}\,\bar{\nu}_{eh}^E - \varepsilon_{\text{ion}}S - w_{\text{rad}} - Q_e, \tag{12}$$

with the electron energy flux

$$\mathbf{h}_e = \frac{5}{2}k_B T_e\,\mathbf{\Gamma}_e - \kappa_e\nabla T_e + k_B T_e\,n_e\Big[C_i(\mathbf{u}_e - \mathbf{u}_i) + C_n(\mathbf{u}_e - \mathbf{u}_n)\Big]. \tag{13}$$

The terms on the right-hand side of equation (12) are Joule heating by the field, elastic energy transfer to the heavy species, the ionization-energy sink, the net radiative loss, and the radial heat loss of the electrons. In the elastic term, $\bar{\nu}_{eh}^E = \bar{\nu}_{en}^E + \bar{\nu}_{ei}^E$, where the superscript E denotes the energy-exchange average of collision frequencies (Appendix B). The ionization sink is the net ionization rate multiplied by $\varepsilon_{\text{ion}} = 15.76$ eV, the first ionization potential of argon. The energy flux (13) is the sum of the enthalpy flux at $\frac{5}{2}k_B T_e$ per electron, Fourier conduction with the zero-current electron thermal conductivity κ_e, and a thermal-diffusion heat flux [39]. The net radiative loss w_{rad} is that of Almeida et al. [11], taken from Benilov and Naidis [38].

A one-dimensional domain has no lateral boundary, so the Joule heat generated away from the electrode can leave only by radiation and by conduction through the two ends of the domain. In a real arc, by contrast, the bulk plasma loses most of its Joule heat radially, by conduction [40]. Without a lateral sink, the one-dimensional column overheats, and its temperature depends linearly on the length of the domain. We therefore represent the radial conduction of a column of radius R by a volumetric loss for each energy channel [40],

$$Q_e = \Lambda\,(\kappa_e + \kappa_R)\,(T_e - T_{\text{edge}}), \qquad Q_h = \Lambda\,\kappa_h\,(T_h - T_{\text{edge}}), \qquad \Lambda = \frac{4}{R^2}, \tag{14}$$

where T_{edge} is the temperature at the edge of the column. Ion pairs also diffuse radially out of the hot core and recombine near the cold edge, releasing their ionization energy there. This heat loss is represented by the reactive thermal conductivity κ_R (Appendix B.4), which is attributed to the electrons [16]. The radius is not prescribed; instead, Λ is chosen so that the bulk-plasma state imposed at $x = L$ (Section 2.3) is a uniform solution of the column, in which Joule heating balances radiation and radial conduction,

$$\frac{J_{\text{app}}^2}{\sigma} = w_{\text{rad}} + \Lambda\,(\kappa_e + \kappa_R + \kappa_h)\,(T_{\text{bulk}} - T_{\text{edge}}), \tag{15}$$

with all quantities evaluated at the bulk state and σ the electrical conductivity of the plasma. Closer to the electrode, the axial gradients develop over 10^{-6}–10^{-4} m, while R is of order 10^{-3} m, so the radial losses are negligible there in comparison.

The common heavy-particle temperature is governed by

$$n_h c_v\left(\frac{\partial T_h}{\partial t} + \mathbf{u}\cdot\nabla T_h\right) = \nabla\cdot(\kappa_h\nabla T_h) + \frac{3}{2}n_e k_B(T_e - T_h)\,\frac{2m_e}{M}\,\bar{\nu}_{eh}^E + Q_{in} - Q_h - n_h k_B T_h\,\nabla\cdot\mathbf{u}, \tag{16}$$

where $c_v = \frac{3}{2}k_B$ is the translational heat capacity per heavy particle and κ_h is the heavy-particle thermal conductivity (Appendix B). The elastic-exchange term is the same as in the electron

energy equation (12), but with the opposite sign. The ionization energy and the radiative loss are charged entirely to the electrons and do not appear here. The term Q_h is the radial heat loss of the heavy particles, equation (14), and the ion–neutral frictional heating is

$$Q_{in} = M n_i \, \bar{\nu}_{in}^{M} \, |\mathbf{u}_i - \mathbf{u}_n|^2, \tag{17}$$

which is positive definite and reduces to the Joule dissipation of the ion drift when the neutral gas is at rest. The last term in equation (16) is the compression work of the heavy gas.

The velocity of the heavy gas as a whole is advanced by an inviscid single-fluid momentum equation,

$$\frac{\partial(\rho\mathbf{u})}{\partial t} + \nabla\cdot(\rho\mathbf{u}\mathbf{u}) = -\nabla p + e(n_i - n_e)\mathbf{E}, \qquad p = n_e k_B T_e + n_h k_B T_h. \tag{18}$$

The state advanced in time is therefore $\{n_e, n_i, n_n, \mathcal{E}, T_h, \rho u, \phi\}$. It is governed by the continuity equations (1) with the fluxes (9), (10) and (8), Poisson's equation (11), the electron and heavy-particle energy equations (12) and (16), and the heavy-gas momentum equation (18).

2.3 Boundary conditions

The system is closed by conditions at the two ends of the domain: the electrode surface, $x = 0$, and the edge of the near-electrode layer, $x = L$, where the solution is matched to the adjacent LTE bulk plasma. The potential is referenced to zero at the electrode, $\phi(0) = 0$.

Electrode surface, $x = 0$. Electrons reaching the surface are absorbed; they are assumed to have a Maxwellian distribution at their local temperature T_e. At the cathode, thermionic emission of electrons, Γ_{em}, with a Maxwellian distribution at the wall temperature T_{wall}, is also included. The electron flux into the surface is

$$\boldsymbol{\Gamma}_e\cdot\hat{\mathbf{n}} = \tfrac{1}{4} n_e \bar{c}_e - \Gamma_{\mathrm{em}}, \qquad \bar{c}_e = \sqrt{\frac{8k_B T_e}{\pi m_e}}. \tag{19}$$

The emitted flux is given by the Richardson–Dushman law with Schottky lowering of the work function W by the field at the surface,

$$\Gamma_{\mathrm{em}} = \frac{A_R T_{\mathrm{wall}}^2}{e} \exp\!\left(-\frac{W - \Delta W}{k_B T_{\mathrm{wall}}}\right), \qquad \Delta W = \sqrt{\frac{e^3 E_w}{4\pi\varepsilon_0}}, \qquad E_w = \max(\mathbf{E}\cdot\hat{\mathbf{n}}, 0), \tag{20}$$

where A_R is the Richardson constant and W the work function. Each absorbed electron removes a mean energy $2k_B T_e$ from the plasma, and each emitted electron brings in $2k_B T_{\mathrm{wall}}$, so the boundary condition for the electron energy equation (12) is

$$\mathbf{h}_e\cdot\hat{\mathbf{n}} = 2k_B T_e \, \tfrac{1}{4} n_e \bar{c}_e - 2k_B T_{\mathrm{wall}} \, \Gamma_{\mathrm{em}}. \tag{21}$$

Ions reaching the surface are assumed to recombine there and to return to the plasma as atoms. The ion flux into the surface is taken as the sum of a thermal-effusion flux and a directed flux. Let $\boldsymbol{\Gamma}_i^d$ denote the ion flux (10) without its density-diffusion term, $\boldsymbol{\Gamma}_i^d = \boldsymbol{\Gamma}_i + [k_B T_h/(M\bar{\nu}_i^M)]\nabla n_i$. Ions are absorbed at their thermal-effusion flux plus this directed flux where it points into the wall,

$$\boldsymbol{\Gamma}_i\cdot\hat{\mathbf{n}} = \tfrac{1}{4} n_i \bar{c}_i + \max\!\left(\boldsymbol{\Gamma}_i^d\cdot\hat{\mathbf{n}}, 0\right), \qquad \bar{c}_i = \sqrt{\frac{8k_B T_{\mathrm{wall}}}{\pi M}}. \tag{22}$$

Where the sheath field drives the ions into the electrode, the directed term dominates. Elsewhere, the thermal-effusion flux ensures that ions reaching the wall are absorbed even where transport alone does not carry them into it.

The wall is impermeable to the heavy gas, $\mathbf{u}\cdot\hat{\mathbf{n}} = 0$; this is the boundary condition for the heavy-gas momentum equation (18). Equation (8) then gives

$$\boldsymbol{\Gamma}_n\cdot\hat{\mathbf{n}} = -\boldsymbol{\Gamma}_i\cdot\hat{\mathbf{n}}\,: \tag{23}$$

every ion neutralized at the surface re-enters the plasma as an atom. Secondary electron emission by ion impact is not included.

The heavy-particle temperature at the surface equals the electrode temperature, $T_h(0) = T_{\mathrm{wall}}$. At the cathode, T_{wall} is obtained by balancing the heat flux q_w delivered by the plasma against steady conduction through the cathode body,

$$q_w = \frac{k_w}{L_w}\left(T_{\mathrm{wall}} - T_0\right), \tag{24}$$

where k_w is the thermal conductivity of the cathode, L_w its length, and T_0 the temperature of its base. This crude heat balance is intended only to capture the trend of the cathode temperature with the current. The anode is usually water-cooled, so its surface temperature is assumed to be constant.

The heat flux into the surface is

$$q_w = (2k_B T_e + W)\,\tfrac{1}{4} n_e \bar{c}_e - (2k_B T_{\mathrm{wall}} + W)\,\Gamma_{\mathrm{em}} + (\varepsilon_{\mathrm{ion}} - W)\,\mathbf{\Gamma}_i \cdot \hat{\mathbf{n}} - \kappa_h \nabla T_h \cdot \hat{\mathbf{n}}. \tag{25}$$

The first term is the energy delivered by the absorbed electrons: their kinetic energy plus the work function released on entering the metal. The second is emission cooling: each emitted electron removes the work function and its kinetic energy. The third is the energy released when an ion recombines at the surface, $\varepsilon_{\mathrm{ion}}$, less the work function spent to extract the neutralizing electron. The last is heat conduction from the heavy gas. At the anode, $\Gamma_{\mathrm{em}} = 0$ and the emission term vanishes.

Bulk-plasma edge, $x = L$. The electron and ion densities and both temperatures are set to Dirichlet values from the adjacent LTE bulk plasma: $n_e = n_e^{\mathrm{bulk}}$, $n_i = n_i^{\mathrm{bulk}}$, and $T_e = T_h = T_{\mathrm{bulk}}$; the last of these is the boundary condition for both energy equations. This bulk state, together with the applied current density J_{app}, also fixes the radial-loss coefficient Λ through equation (15), so that the imposed values are consistent with the energy balance of the column. The neutral density follows from Dalton's law at the background pressure p_0,

$$n_n = \frac{p_0 - n_e k_B T_e}{k_B T_h} - n_i. \tag{26}$$

This edge is open to the heavy gas: the heavy-gas momentum equation (18) sees the bulk state, and hence the pressure p_0, outside the boundary, and the heavy-gas velocity is given a zero normal gradient. The arc current is imposed through the electric field in Poisson's equation (11): the total current density at the boundary, conduction plus displacement, is held equal to the applied current density J_{app},

$$e\,(\mathbf{\Gamma}_i - \mathbf{\Gamma}_e) + \varepsilon_0 \frac{\partial E}{\partial t} = J_{\mathrm{app}}, \tag{27}$$

which is integrated in time as an evolution equation for the boundary field.

2.4 Numerical implementation

The model is solved with a finite-volume code built on the block-structured adaptive mesh refinement framework AMReX [41]. The domain $0 \le x \le L$ is divided into cells, and the cell averages of n_e, n_i, n_n, $\mathcal{E}$, T_h, ρu and ϕ are the discrete unknowns. Every conservation law is written in flux form, with the fluxes evaluated on the cell faces. The species fluxes (9) and (10) are split into a density-diffusion part, $-D\,\partial n/\partial x$, and a non-diffusive part, nv, where the velocity v collects the drift in the field, the temperature-gradient and thermal-force terms, the mixture force and the collisional drag. In the sheath, the drift and diffusion fluxes of the electrons nearly cancel, and their sum is two to three orders of magnitude smaller than either. A small relative error in either part would therefore be a large error in the net flux. The face flux is accordingly the exponentially fitted (Scharfetter–Gummel) flux [42],

$$\Gamma_f = \frac{D_f}{\Delta x}\Big[B(-\psi)\,n_{\mathrm{L}} - B(\psi)\,n_{\mathrm{R}}\Big], \qquad \psi = \frac{v_f\,\Delta x}{D_f}, \qquad B(\psi) = \frac{\psi}{e^{\psi} - 1}, \tag{28}$$

where L and R denote the cells on either side of the face and ψ is the cell Péclet number. Equation (28) is exact for the exponential profile of a Boltzmann balance; it reduces to central differencing as $\psi \to 0$ and to upwinding as $|\psi| \to \infty$. The enthalpy part of the electron energy flux (13) is computed from this discrete electron flux, $\frac{5}{2} k_B T_e\,\Gamma_{e,f}$, so that each electron crossing a face carries exactly its own enthalpy. Heat conduction in both energy equations and Poisson's equation (11) are discretized with central differences, and the face field $E_f = -(\phi_{\mathrm{R}} - \phi_{\mathrm{L}})/\Delta x$ is used in every face flux and in the Joule heating. The heavy-gas momentum equation (18) is discretized with a Roe-type flux [43]. At the electrode, the wall fluxes (19)–(23) are imposed as the face fluxes, evaluated with the densities and temperatures of the cell adjacent to the wall; at $x = L$, the Dirichlet values of Section 2.3 are imposed on the boundary face.

All cases in this study use a uniform base mesh of 512 cells with four levels of refinement towards the electrode, each with a refinement ratio of 4. The refined regions are nested layers adjacent to the electrode, 1.0×10^{-4}, 2.0×10^{-6}, 7.3×10^{-7} and 2.7×10^{-7} m deep, with cell sizes of 244, 61, 15.3 and 3.8 nm, so that the mesh has 944 cells in total. These sizes are chosen to

resolve the length scales estimated in Section 2.1, down to a fraction of the Debye length, at both the cathode and the anode. The levels are coupled conservatively. The equations are advanced in time, from a uniform initial state at the bulk-plasma conditions or from a neighboring steady state, until a steady state is reached. All equations, including Poisson's equation and the galvanostatic condition (27), are solved together at each time step. A run is considered steady when, for each of ϕ, n_e, n_i, $\mathcal{E}$ and T_h, the relative change over a time step, $\|\mathbf{U}^{n+1} - \mathbf{U}^n\|_2/\|\mathbf{U}^{n+1}\|_2$, extrapolated to 1 μs of simulated time, stays below 10^{-5} on several consecutive checks. The steady states are checked a posteriori by the residual of the electron energy equation and by the uniformity of the total current density (Section 4).

3 Validation

The model is validated against two published unified models of the near-electrode layers of atmospheric-pressure argon arcs: the near-cathode layer computed by Almeida et al. [11] and the near-anode layer computed by Semenov et al. [13]. Both works solve the multi-fluid equations for electrons, ions and atoms, with separate electron and heavy-particle temperatures, coupled to Poisson's equation. The two models differ, however, in geometry. Semenov et al. use a planar one-dimensional model, as the present one does, so the current density is the same at every x. Almeida et al. instead solve their equations in spherical symmetry about a hemispherical cathode of radius $r_c = 1$ mm, so their current density decreases with distance as $j_c\, r_c^2/(r_c + x)^2$, whereas the present model keeps $j = j_c$ throughout. At the outer boundary of their domains, which is 1 cm from the electrode, both assume that the plasma is uniform and in LTE, with Joule heating balanced by radiation alone. Their solutions show, however, that the plasma is already close to LTE much nearer the electrode: at $x = 5 \times 10^{-4}$ m, T_e and T_h differ by less than 0.3 % in both published solutions, and the charged-particle density is within 2 % of the Saha density, although the plasma there is not yet uniform. We therefore restrict the present domain to this length and take the boundary condition at $x = L$ from the published solutions.

The conditions of the published calculations were reproduced as closely as possible. The electrode surface temperature was prescribed at the value used by Almeida et al. and Semenov et al., respectively, rather than obtained from the heat balance (24), and the radial loss (14) was switched off, $\Lambda = 0$, since neither model includes it. The domain was discretized as described in Section 2.4. Throughout the space-charge sheath, the region where $|n_i - n_e|/n_i > 1\,\%$ ($x \lesssim 1.1$ μm at the cathode and 0.7 μm at the anode), the cells are smaller than the local electron Debye length, by a factor of at least 1.5 at the cathode and 6 at the anode. The equations were integrated in time at a fixed current density to a steady state.

3.1 Near-cathode layer

Almeida et al. [11] computed the near-cathode layer of an atmospheric-pressure argon arc for three current densities, and concluded that only one of these cases is realizable in practice: a tungsten cathode with a work function of 4.55 eV, thermionic emission (20) included, a surface temperature of 3500 K and a current density of 10^7 A m^{-2}. We therefore model this case; the two solutions are compared in figure 1. The temperature profiles agree closely across the whole layer (figure 1(b)). The electron temperature stays near 17.5–18 kK through the sheath and the inner part of the ionization layer and decreases towards the bulk plasma, while the heavy-particle temperature rises from the wall value to a maximum of 13.9 kK at $x \approx 2 \times 10^{-4}$ m, compared with 13.6 kK in the solution of Almeida et al. From $x \approx 3 \times 10^{-6}$ m outwards, both temperatures agree with theirs within 0.4 kK (3 %).

The ion fraction of the current, figure 1(c), is reproduced over the whole layer. At the cathode surface the present model gives $j_i/j = 0.218$, compared with 0.216 reported by Almeida et al.; j_i/j stays at this value through the sheath, falls to zero at $x \approx 1.5 \times 10^{-4}$ m, and becomes slightly negative further out, where the ion flux is directed away from the cathode. Almeida et al. also place the sheath edge where the charge separation reaches 1 %. The present model gives a sheath voltage of 9.96 V at $x_s = 1.1$ μm, compared with 10.2 V at $x_s = 1.5$ μm in their solution. In the ionization layer, $10^{-5} \lesssim x \lesssim 3 \times 10^{-4}$ m, the charged-particle densities of the present model are 7–15 % higher than theirs (figure 1(a)). The largest differences are in the space-charge sheath, $x \lesssim 10^{-6}$ m. At $x = 10^{-7}$ m, the electron and ion densities of the present model are higher by a factor of about 2.5, and the field strength is 2.3×10^7 V m^{-1}, compared with 1.6×10^7 V m^{-1}.

3.2 Near-anode layer

The anode case is that of figure 3(b) of Semenov et al. [13]: a non-emitting anode, $\Gamma_{\mathrm{em}} = 0$, with surface temperature 1000 K and current density 1.5×10^7 A m^{-2}. The comparison is shown in

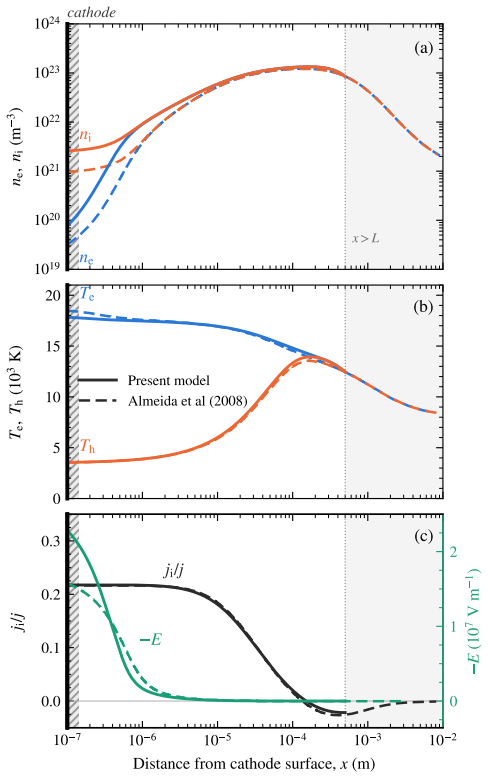


Figure 1. Near-cathode layer in argon at atmospheric pressure, cathode surface temperature $T_w = 3500$ K and current density $j_c = 10^7$ A m^{-2}: present model (solid lines) and Almeida et al. [11] (dashed lines, figures 5(b) and 6). (a) Electron and ion number densities. (b) Electron and heavy-particle temperatures. (c) Ion fraction of the current density, j_i/j (left axis), and field strength $-E$ (right axis). The hatched strip at the left marks the cathode surface. The shaded region, $x > L$, lies outside the computational domain.

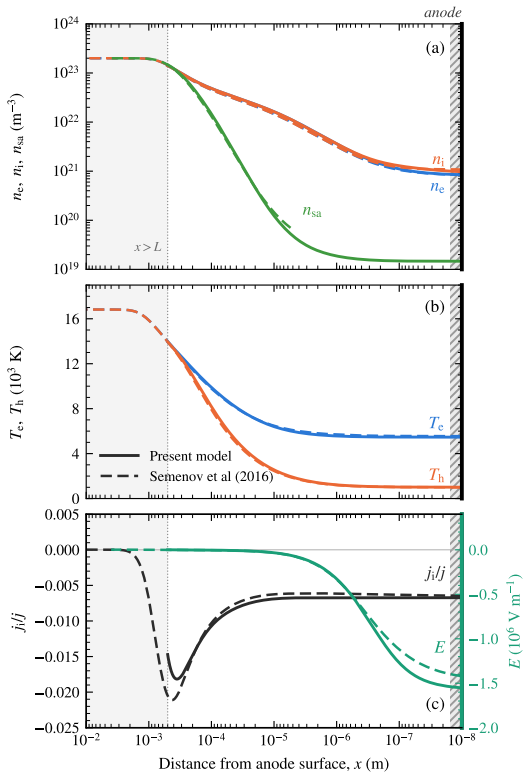


Figure 2. Near-anode layer in argon at atmospheric pressure, anode surface temperature $T_s = 1000$ K and current density $j = 1.5 \times 10^7$ A m^{-2}: present model (solid lines) and Semenov et al. [13] (dashed lines, figures 3(b), 4 and 5). (a) Electron and ion number densities and the Saha density n_{sa}. (b) Electron and heavy-particle temperatures. (c) Ion fraction of the current density, j_i/j (left axis), and electric field E (right axis). The distance axis is reversed, so that the anode surface, marked by the hatched strip, is at the right. The shaded region, $x > L$, lies outside the computational domain.

figure 2. The electron and ion densities agree with those of Semenov et al. within 15 % across the layer (figure 2(a)). Following Semenov et al., the figure also shows the Saha density n_{sa}, the charged-particle density that the local T_e and T_h would give in ionization equilibrium with quasi-neutrality at the pressure p_0. Near the anode the actual densities exceed n_{sa} by nearly two orders of magnitude. The anode region is therefore far from ionization equilibrium: charged particles are supplied by diffusion from the hotter plasma rather than produced locally, and both models capture this departure.

The temperature profiles agree closely (figure 2(b)). At the anode surface, the electron temperature is 5.46 kK, compared with 5.59 kK reported by Semenov et al., and the heavy-particle temperatures differ by less than 0.3 kK across the layer. The electric field near the anode, figure 2(c), is negative: it is directed towards the anode, so that the anode lies at a lower potential than the adjacent plasma and the electrons are retarded. The field at the surface is -1.57×10^6 V m^{-1}, compared with -1.42×10^6 V m^{-1} in the solution of Semenov et al. Ions flow into the anode, so j_i/j is negative there. At the surface, the present model gives $j_i/j = -0.0067$, compared with their -0.0065. With the same definition of the sheath edge as at the cathode, the present model gives $x_s = 0.70$ μm and $\phi_s = 0.58$ V, compared with 0.71 μm and 0.54 V obtained from the density and field profiles of Semenov et al.

The main difference is in the minimum of j_i/j, which is -0.018 at $x = 3.5 \times 10^{-4}$ m in the present model and -0.021 at $x = 4.3 \times 10^{-4}$ m in the solution of Semenov et al. This difference reflects the different models. Imposing their state at $x = L$ fixes the values of the densities and temperatures there, but not their gradients, which follow from each model's governing equations, and j_i/j depends on these gradients.

The present model reproduces the structure of the near-electrode layer for both cathode and

anode cases: the temperatures, the current partition between ions and electrons, the sign and magnitude of the near-electrode field, the sheath voltages (within 2.5 % at the cathode and 7 % at the anode) and the departure from ionization equilibrium. Apart from the charged-particle densities in the cathode sheath and the minimum of j_i/j near the outer boundary of the anode calculation, the temperatures agree within 3–5 % and the charged-particle densities within 15 %. The differences most likely arise from the different descriptions of the transport properties (Appendix B) and of the ion boundary condition (22).

4 Results

The model is now applied to the cathode and anode regions of a high-intensity arc in argon at atmospheric pressure, with an arc current of 100–200 A. Experiments and LTE models of such arcs [35, 10, 37, 8, 9, 32, 33] give the state of the plasma near the electrodes. Together with the scale estimates of Section 2.1 and previous studies [11, 23, 17, 24, 34, 13], a distance of 5×10^{-4} m from the electrodes was considered to be a reasonable choice for the existence of LTE conditions. At this distance, $x = L$, we take $T_{\text{bulk}} = 21\,000$ K and $J_{\text{app}} = 10^8$ A m^{-2} for the cathode, and $T_{\text{bulk}} = 14\,000$ K and $|J_{\text{app}}| = 10^7$ A m^{-2} for the anode (table 2). Unlike in the validation cases of Section 3, the radial loss (14) is included, with the reactive conductivity in the electron channel. The calibration (15) gives a column radius $R = 3.4$ mm for the anode, which is typical of a diffuse anode attachment [10, 16, 32], and 0.51 mm for the cathode, which is typical of the radius of a cathode spot [10, 37]. At 14 000 K, $\kappa_R = 0.93$ W m^{-1} K^{-1} is comparable to $\kappa_e = 1.07$ W m^{-1} K^{-1}, whereas at 21 000 K the plasma is almost fully ionized and $\kappa_R = 0.03$ W m^{-1} K^{-1} is negligible. At $x = L$, all densities and temperatures are Dirichlet values (Section 2.3). The electrode materials used in the LTE results are copper for the anode and thoriated tungsten for the cathode; these materials are therefore adopted here. The anode has a work function of 4.65 eV and is maintained at a uniform prescribed surface temperature of $T_{\text{wall}} = 1000$ K. The cathode surface temperature is determined by the heat balance in Eq. (24). The cathode is assumed to have uniform properties $k_w = 100$ W m^{-1} K^{-1}, $L_w = 1$ cm, $T_0 = 1000$ K, $A_R = 3 \times 10^4$ A m^{-2} K^{-2}, and $W = 2.63$ eV. The computational mesh is identical to that described in Section 2.4. For both steady-state solutions, the residual of the electron energy equation, Eq. (12), quantified by $\partial\mathcal{E}/\partial t$, remains below 1.2×10^{-3} of the largest-magnitude term in the equation throughout the domain. The total current density is uniform to within a relative variation of 10^{-8}.

Table 2. Conditions (top) and computed electrode quantities (bottom). The sheath edge x_s is the distance from the electrode at which $|n_i - n_e|/n_i$ falls to 1 %, and ϕ_s is the potential there; $\phi(L)$ is the potential at the bulk-plasma boundary. Both potentials are relative to the electrode. Fluxes are towards the electrode.

	Cathode	Anode
Current density $\lvert J_{\text{app}}\rvert$ (A m^{-2})	10^8	10^7
Bulk temperature T_{bulk} (K)	21 000	14 000
Bulk electron density n_e^{bulk} (m^{-3})	1.7×10^{23}	1.5×10^{23}
Thermionic emission	on	off
Surface temperature T_{wall} (K)	3670	1000 (prescribed)
T_e at the surface (K)	18 200	5100
n_e, n_i at the surface (m^{-3})	2.1×10^{21}, 6.5×10^{21}	5.7×10^{20}, 6.7×10^{20}
Sheath edge, x_s (m)	5.8×10^{-7}	8.0×10^{-7}
ϕ_s, $\phi(L)$ (V)	3.6, 12.5	0.60, 3.43
Field at the surface (V m^{-1})	1.8×10^7	-1.5×10^6
j_i/j at the surface	0.054	−0.0070
Absorbed electron flux $\frac{1}{4} n_e \bar{c}_e$ (m^{-2} s^{-1})	4.4×10^{26}	6.3×10^{25}
Emitted electron flux Γ_{em} (m^{-2} s^{-1})	1.03×10^{27}	–
Ion flux $\boldsymbol{\Gamma}_i \cdot \hat{\mathbf{n}}$ (m^{-2} s^{-1})	3.3×10^{25}	4.4×10^{23}
Heat flux q_w (MW m^{-2})	26.7	65.7

4.1 Structure of the near-electrode layers

Cathode. A space-charge sheath 0.58 μm thick forms at the surface. In it, n_i exceeds n_e, by up to a factor of 3.1 at the surface (figure 3(a)), and across it the potential rises by 3.6 V. The field at the surface is 1.8×10^7 V m^{-1}. Beyond the sheath, the potential keeps rising through the quasi-neutral ionization layer, to 8.3 V at $x = 10^{-4}$ m (figure 3(e)). Further out, the plasma approaches the bulk state, and the remaining rise to 12.5 V at $x = L$ is mostly the ohmic drop of the column, whose field is 9.5×10^3 V m^{-1}. The electron temperature is nearly uniform across the whole layer, 18.2–21 kK, while the heavy-particle temperature rises from the surface value of 3.7 kK and approaches T_e only at $x \approx 10^{-4}$ m (figure 3(c)). Over the same distance, the plasma becomes almost fully ionized.

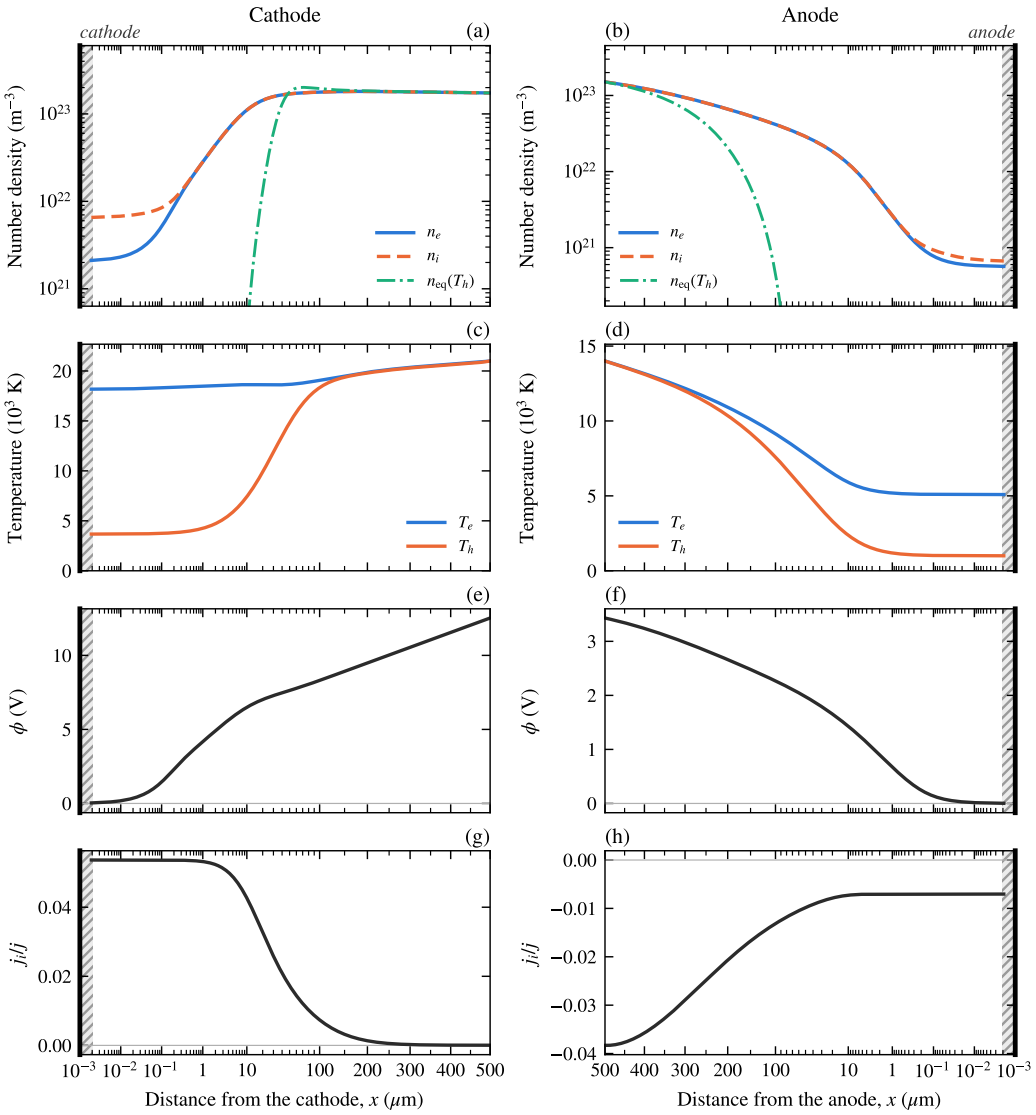


Figure 3. Steady profiles at the cathode (left) and the anode (right). x is the distance from the electrode; the cathode surface is at the left edge of its panels and the anode surface at the right edge of its panels, marked by the hatched strip. The distance axis is scaled as $\ln x + x/x_c$, with $x_c = 50$ μm: logarithmic near the electrode and smoothly becoming linear towards the bulk plasma, so that both the sheath and the approach to the bulk are visible. (a), (b) Electron and ion number densities, and the equilibrium density $n_{eq}(T_h)$: the charged-particle density at which ionization and recombination, equation (2), balance at $T_e = T_h$ equal to the local heavy-particle temperature, with quasi-neutrality and Dalton's law at p_0. At $x = L$ it is the imposed bulk density. (c), (d) Electron and heavy-particle temperatures. (e), (f) Potential relative to the electrode. (g), (h) Ion fraction of the current density, positive for ions flowing into the electrode.

Close to the cathode, the charged-particle density is far above the equilibrium density $n_{\mathrm{eq}}(T_h)$ that the reactions would give at the local heavy-particle temperature (figure 3(a)), by a factor of 400 at 10 μm. The two densities come within a factor of 2 of each other beyond 33 μm, and within 1 % only beyond 0.2 mm, where T_h has approached T_e. The emitted electron flux, 1.03×10^{27} m^{-2} s^{-1}, corresponds to a current density of 1.65×10^{8} A m^{-2}, 1.65 times the arc current density. It is partly offset by the plasma electrons that reach the surface against the sheath field, 4.4×10^{26} m^{-2} s^{-1} (table 2), and the ions carry 5.4 % of the current at the surface (figure 3(g)).

Anode. The space-charge sheath is practically absent. The charge separation exceeds 1 % only within 0.80 μm of the surface, across which the potential changes by only 0.60 V, and at the surface n_i exceeds n_e by 18 % (figure 3(b)). The anode lies 3.43 V below the bulk-plasma boundary, and this drop is distributed over the whole quasi-neutral layer (figure 3(f)): 1.43 V within 10 μm and 2.27 V within 100 μm. The field is directed towards the anode and retards the electrons, which reach the surface by diffusion down the steep density gradient: across the layer, the charged-particle density falls by a factor of 270, from 1.5×10^{23} to 5.7×10^{20} m^{-3}. These densities exceed the equilibrium density $n_{\mathrm{eq}}(T_h)$ everywhere (figure 3(b)): by a factor of 90 at 100 μm, 3.2 at 200 μm and 1.4 at 300 μm, while closer to the anode $n_{\mathrm{eq}}(T_h)$ is negligible. The anode layer is over-populated with charged particles, which recombine, and n_e is within 10 % of $n_{\mathrm{eq}}(T_h)$ only within 0.1 mm of $x = L$. The electron and heavy-particle temperatures decrease together towards the anode, but T_h falls faster, to the prescribed surface temperature of 1 kK, so that the two separate progressively: by 1.6 kK at $x = 10^{-4}$ m, 3.5 kK at $x = 10^{-5}$ m and 4.1 kK at the surface, where $T_e = 5.1$ kK (figure 3(d)). The ions flow into the anode, $j_i/j < 0$, and carry only 0.7 % of the current at the surface (figure 3(h)).

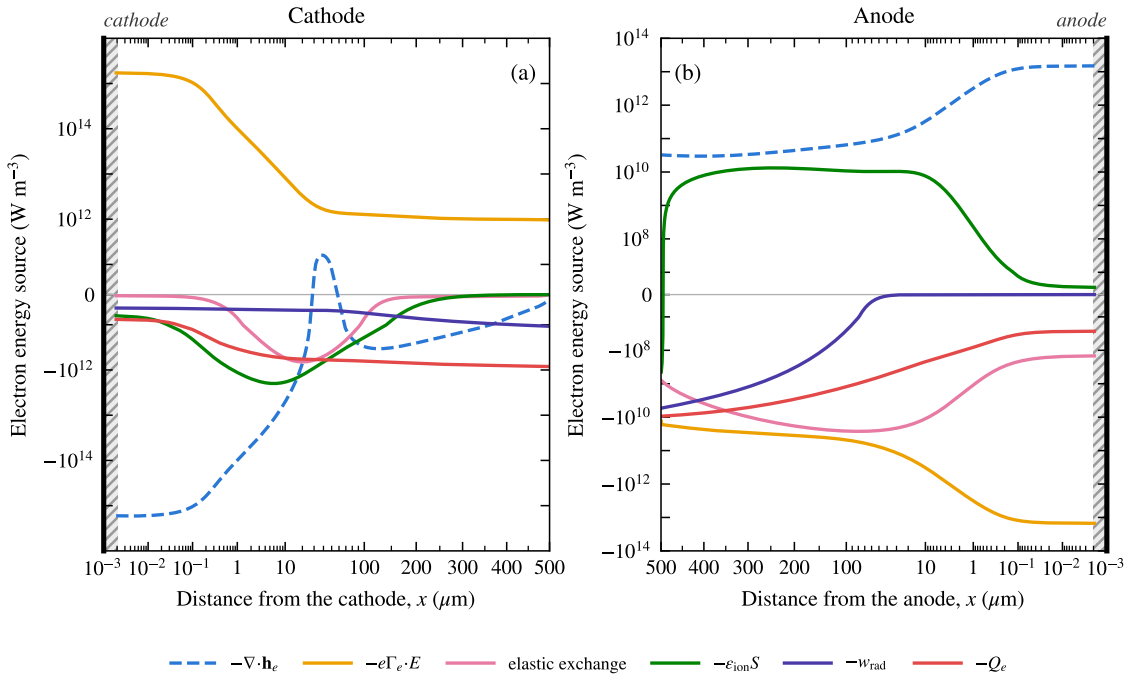


Figure 4. Steady electron energy balance, equation (12), for (a) the cathode and (b) the anode: every term except $\partial\mathcal{E}/\partial t$. Dashed: the divergence of the electron energy flux (13), $-\nabla\cdot\mathbf{h}_e$. Solid: the volume terms, i.e. Joule heating, elastic exchange with the heavy particles, the ionization sink and the radiative loss. Positive terms heat the electrons. The hatched strip marks the electrode surface: the cathode at the left of panel (a), the anode at the right of panel (b). The distance axis is scaled as $\ln x + x/x_c$, with $x_c = 50$ μm: logarithmic near the electrode and smoothly becoming linear towards the bulk plasma, so that both the sheath and the approach to the bulk are visible. The vertical axis is logarithmic in $|\cdot|$ above 10^{11} (a) and 10^{7} W m^{-3} (b), and linear below.

4.2 Electron energy balance

Figure 4 shows every term of the steady electron energy equation (12). The divergence of the electron energy flux (13), $-\nabla\cdot\mathbf{h}_e$, is plotted as one term: it combines the enthalpy flux, conduction and the thermal-diffusion heat flux. The volume terms are Joule heating $-e\boldsymbol{\Gamma}_e\cdot\mathbf{E}$, elastic exchange with the heavy particles, the ionization term $-\varepsilon_{\mathrm{ion}}S$, radiation $-w_{\mathrm{rad}}$ and the radial loss $-Q_e$ (14). Positive terms heat the electrons, and at steady state the terms sum to zero. Integrated over x,

each term gives a power per unit area, quoted below in $\mathrm{MW\,m^{-2}}$. The balance is very different at the two electrodes. At the cathode, Joule heating is the source, and the electron energy flux carries most of it away. At the anode, the electron energy flux is the source, and the Joule term is a sink.

Cathode. Joule heating is the only source of electron energy, and it is the largest term everywhere (figure 4(a)). It peaks at the surface, 1.7×10^{15} $\mathrm{W\,m^{-3}}$, where the sheath field accelerates the emitted electrons, and falls by more than two orders of magnitude within 10 μm. Over this distance it is balanced almost entirely by $-\nabla \cdot \mathbf{h}_e$, which takes up 590 of the 615 $\mathrm{MW\,m^{-2}}$ released; in the sheath alone, the two are 340 and -339 $\mathrm{MW\,m^{-2}}$. The volume losses there are three to four orders of magnitude smaller. The Joule heat released near the surface is therefore not lost locally: the electron energy flux carries it to the cathode and towards the bulk. Beyond 10 μm, $-\nabla \cdot \mathbf{h}_e$ is small, and the Joule heating, now 10^{12}–10^{13} $\mathrm{W\,m^{-3}}$, is balanced by the volume terms. Ionization is the largest of them out to 100 μm, with a peak of 2.0×10^{12} $\mathrm{W\,m^{-3}}$ at 6 μm. The elastic exchange peaks at 6.6×10^{11} $\mathrm{W\,m^{-3}}$ at 20 μm, where it heats the heavy particles (figure 3(c)). Both terms become negligible beyond 200 μm, where T_h has approached T_e and the plasma is close to ionization equilibrium. Between 28 and 58 μm, $-\nabla \cdot \mathbf{h}_e$ is positive, up to 1.6×10^{11} $\mathrm{W\,m^{-3}}$. There, T_e has a shallow minimum, 18.60 kK at 30 μm, between a maximum of 18.63 kK at 11 μm and the rise towards the bulk, and the electron energy flux converges into it. In the column part of the domain, $x > 100$ μm, the radial loss and, to a lesser extent, radiation take up the Joule heating. At $x = L$, they balance it to within 0.4 % (8.4×10^{11} and 1.1×10^{11} against 9.5×10^{11} $\mathrm{W\,m^{-3}}$), as the calibration (15) requires, and $-\nabla \cdot \mathbf{h}_e$ falls almost to zero. Over the whole domain, Joule heating supplies 1210 $\mathrm{MW\,m^{-2}}$, of which 686 $\mathrm{MW\,m^{-2}}$ leave the domain as electron energy flux: 569 $\mathrm{MW\,m^{-2}}$ into the column at $x = L$, and 117 $\mathrm{MW\,m^{-2}}$ into the cathode with the absorbed plasma electrons, net of the energy of the emitted electrons. The radial loss takes 362 $\mathrm{MW\,m^{-2}}$, most of it at $x > 100$ μm, and ionization 85 $\mathrm{MW\,m^{-2}}$, two thirds of it between 10 and 100 μm. Radiation and elastic exchange take 41 and 40 $\mathrm{MW\,m^{-2}}$.

Anode. The Joule term is a loss throughout the domain (figure 4(b)), because the electrons flow towards the anode against the retarding field. Its magnitude is largest at the surface, 1.5×10^{13} $\mathrm{W\,m^{-3}}$. Even at $x = L$ it is still negative, -1.3×10^{10} $\mathrm{W\,m^{-3}}$, whereas the ohmic heating of the column at the imposed bulk state is $J^2/\sigma = +1.6 \times 10^{10}$ $\mathrm{W\,m^{-3}}$: the retarding field of the anode layer extends over the whole domain. The only source of electron energy is $-\nabla \cdot \mathbf{h}_e$, which is positive everywhere, so the electron energy flux from the column converges over the whole domain. Of the 48.2 $\mathrm{MW\,m^{-2}}$ that enter at $x = L$, only 8.8 $\mathrm{MW\,m^{-2}}$ reach the surface. Within 10 μm of the anode, $-\nabla \cdot \mathbf{h}_e$ and the Joule loss balance each other to within 0.1 % (14.4 $\mathrm{MW\,m^{-2}}$ each), and at the surface the volume terms are five orders of magnitude smaller. Further from the anode, the Joule loss decreases to a few 10^{10} $\mathrm{W\,m^{-3}}$, and the volume terms become comparable to it. The elastic loss to the heavy particles peaks at 2.7×10^{10} $\mathrm{W\,m^{-3}}$ at 79 μm, where $T_e - T_h$ is still 1.6 kK. The ionization term heats the electrons, with a peak of 1.3×10^{10} $\mathrm{W\,m^{-3}}$ at 243 μm: the plasma recombines, $S < 0$, because the charged particles are supplied by diffusion at densities well above the local ionization equilibrium (Section 3.2), and the recombination energy is returned to the electrons. Radiation and the radial loss grow towards the bulk, and at $x = L$ they reach 5.4×10^9 and 9.4×10^9 $\mathrm{W\,m^{-3}}$. Over the whole domain, the 39.4 $\mathrm{MW\,m^{-2}}$ supplied by $-\nabla \cdot \mathbf{h}_e$ and the 4.9 $\mathrm{MW\,m^{-2}}$ of recombination heating cover the Joule loss (34.8 $\mathrm{MW\,m^{-2}}$), the elastic loss (6.7 $\mathrm{MW\,m^{-2}}$), the radial loss (2.2 $\mathrm{MW\,m^{-2}}$) and radiation (0.7 $\mathrm{MW\,m^{-2}}$). Within 5 μm of $x = L$, the ionization term changes sign. This is a property of the boundary condition there: the densities are held at $n_{\mathrm{eq}}(T_{\mathrm{bulk}})$, while T_e in the adjacent cells exceeds T_{bulk} by up to 3 K, so that the plasma is slightly below equilibrium and ionizes. The resulting sink is at most 4.4×10^8 $\mathrm{W\,m^{-3}}$, 3 % of the peak recombination heating.

4.3 Momentum balance and collisionality of the sheaths

The fluxes (9) and (10) follow from the force balances (3) and (4), in which the inertia of each species relative to the heavy gas is neglected. Figure 5 shows the terms of these balances in the two steady states near the electrode: the electric force, the pressure gradient, the two collisional drags combined and the thermal force. In the steady states the heavy gas is at rest ($|\mathbf{u}| \leq 3 \times 10^{-4}$ $\mathrm{m\,s^{-1}}$), so the share $Y_s\mathbf{F}$ of the net force is negligible (below 2×10^{-4} of the largest term). For comparison, the figure also shows the inertia $m_s n_s (\mathbf{u}_s \cdot \nabla)\mathbf{u}_s$ that the model neglects, evaluated a posteriori from the computed fluxes. The retained terms sum to zero to within 3×10^{-3} of the largest one.

Ions. In both sheaths, the electric force on the ions is balanced by the drag (figure 5(a), (b)). At the cathode, the pressure gradient adds at most 23 % to the electric force in the sheath, and 8 % at the surface. At the anode, the field is weaker, and the pressure gradient, which also drives the

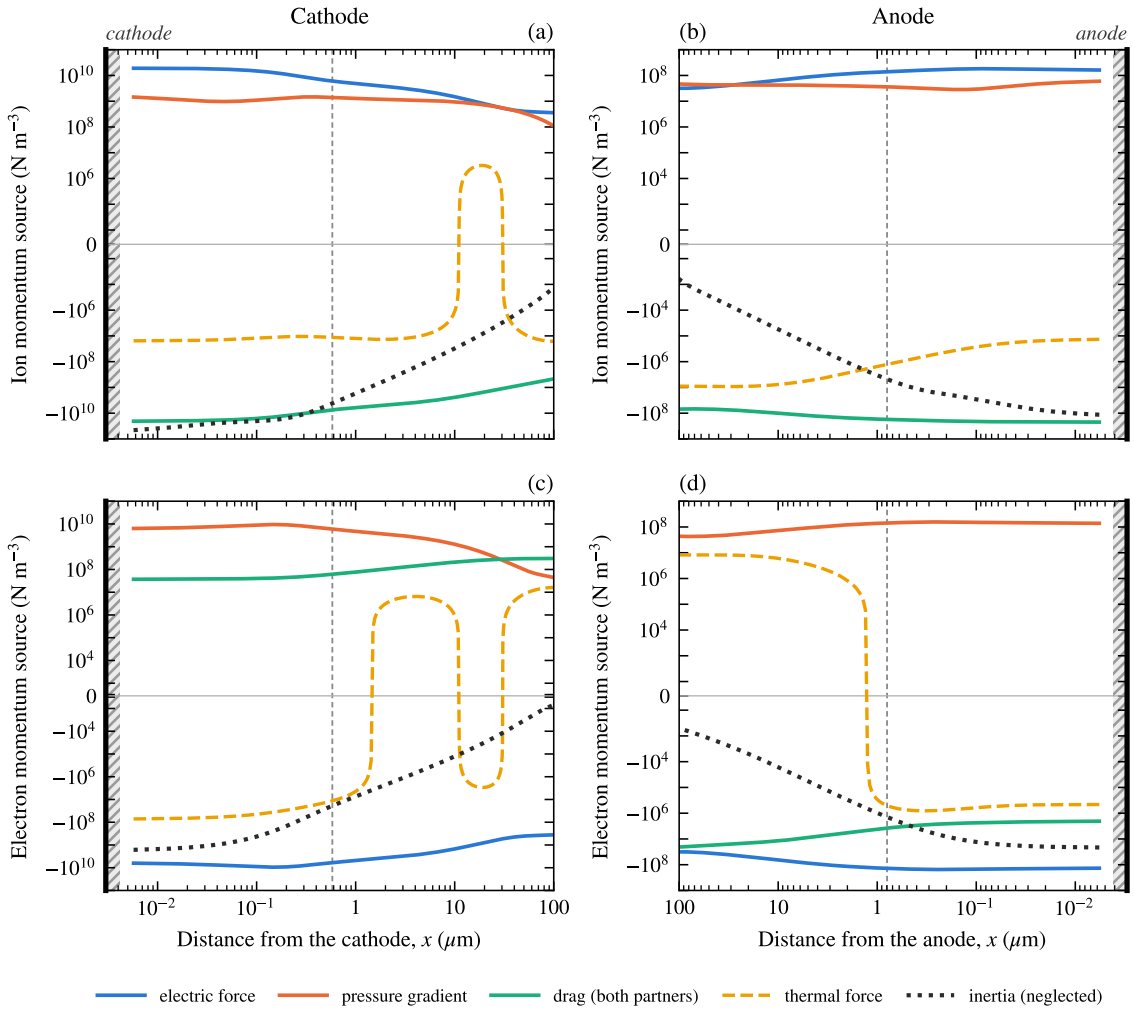


Figure 5. Terms of the steady momentum balances of the ions, equation (4), (a) and (b), and of the electrons, equation (3), (c) and (d), at the cathode (left) and the anode (right), within 100 μm of the electrode. Positive terms push the species towards the electrode. Drag: the friction with both collision partners combined; thermal force: the term in C_e of equation (3) and its reaction C_i on the ions. Dotted: the inertia $-m_s n_s(\mathbf{u}_s \cdot \nabla)\mathbf{u}_s$, which the model neglects, evaluated with $\mathbf{u}_s = \mathbf{\Gamma}_s/n_s$ from the computed fluxes. The vertical dashed line marks the sheath edge x_s (table 2). The vertical axis is logarithmic in $|\cdot|$ and linear within six decades of the largest term.

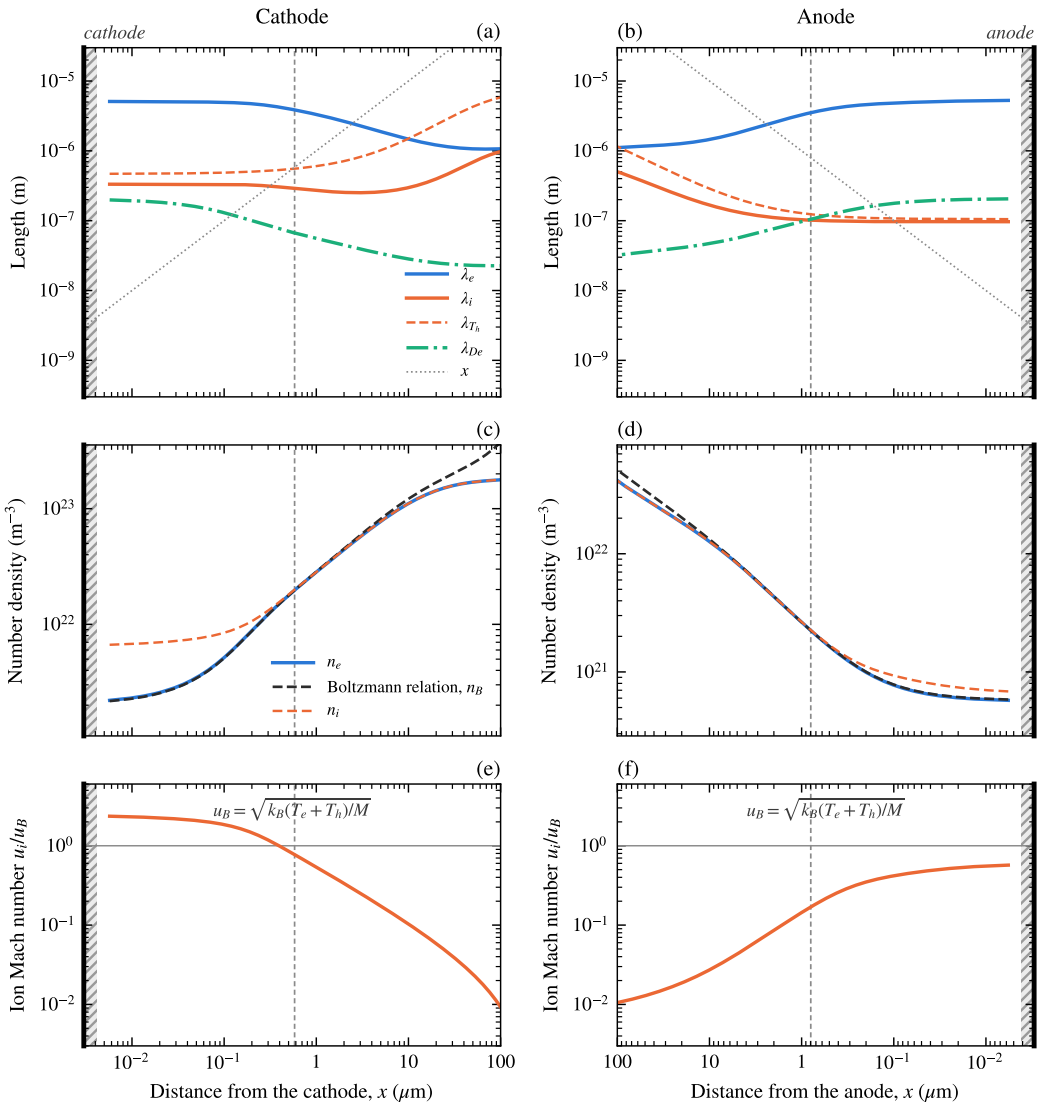


Figure 6. The sheaths at the cathode (left) and the anode (right). (a), (b) Electron and ion mean free paths, the length λ_{T_h} over which ions and atoms exchange energy, and the electron Debye length, from the local state; dotted: the distance x from the electrode. (c), (d) Electron density and the Boltzmann relation $n_B = n_e(x_s)\exp[(e/k_B)\int_{\phi_s}^{\phi} d\phi'/T_e]$ started at the sheath edge, with the ion density for reference. (e), (f) Ion velocity towards the electrode, $u_i = \Gamma_i/n_i$, relative to the Bohm velocity $u_B = \sqrt{k_B(T_e + T_h)/M}$. The vertical dashed line marks the sheath edge x_s.

ions towards the surface, is 37 % of the electric force there. The thermal force on the ions is negligible, below 1 % of the drag in either sheath. The neglected inertia is not small inside the cathode sheath: it exceeds the drag within 0.33 μm of the surface, by a factor of 2.2 at the surface, and stays above 10 % of the drag out to 2.1 μm; beyond 12 μm it is below 1 %. At the anode, the inertia is at most 52 % of the drag (at the surface), and below 10 % beyond 0.17 μm.

Electrons. In both sheaths the field retards the electrons, and the electric force on them is balanced by their pressure gradient (figure 5(c), (d)). In the sheath the two cancel to within 0.8 % (cathode) and 3.1 % (anode) of the electric force. The drag (0.6 and 1.5 % of the electric force at the surface) and the thermal force (1.1 and 0.3 %) are small there. The drag becomes important only in the quasi-neutral layer, where it carries the current: at 10 μm from the surface it is 14 % (cathode) and 18 % (anode) of the electric force. The inertia of the electrons exceeds their drag within 0.34 μm of the cathode (by up to a factor of 44) and within 0.45 μm of the anode (by up to a factor of 10), but it remains smaller than the electric force: at the surface it is a quarter of the electric force at the cathode and 16 % of it at the anode, and at the sheath edge 0.3 and 1.0 %. At the cathode, this inertia is that of the emitted electrons, accelerated by the sheath field.

Mean free paths. Figure 6(a), (b) shows, evaluated with the local densities and temperatures, the mean free paths λ_e and λ_i of table 1, the length $\lambda_{T_h} = 1/[(n_i + n_n)\overline{Q}_{in}]$ over which ions and atoms exchange energy, and the electron Debye length. The electron mean free path is of order microns: 3.9 μm at the cathode sheath edge and 3.6 μm at the anode one. The sheaths, 0.58 and 0.80 μm thick, therefore contain only 0.13 and 0.18 electron mean free paths, and λ_e exceeds the distance from the surface within 2.3 μm of either electrode. The ions are more collisional. At the sheath edge, $\lambda_i = 0.29$ μm at the cathode and 0.10 μm at the anode, so that the ions cross 1.8 and 7.7 mean free paths in the sheaths (1.1 and 6.7 counting only the collisions with atoms). The anode sheath is thus collision-dominated for the ions, consistent with their small inertia. The cathode sheath is transitional: the ions make one to two collisions across it, and their inertia exceeds the drag over its inner half. There the ions cannot follow the steep rise of the field, and the local relation between field and ion velocity in equation (10) overestimates the ion velocity near the surface. The electrons cross both sheaths practically without collisions. Their balance there is, however, not one of collisional transport: the drag is a few per cent of the electric force, and the field and the pressure gradient balance each other as in the Boltzmann equilibrium of a Maxwellian population retarded by a potential barrier, which does not require collisions. The electron description therefore remains adequate in the sheaths, except within about 0.1 μm of the cathode, where the inertia of the emitted electrons is no longer small. Beyond 2–3 μm from either electrode, the inertia of both species is below 10 % of the drag, and the inertialess, collision-dominated description of Section 2.2 holds.

A single heavy-particle temperature. The model describes the ions and the atoms by one temperature T_h. This requires, first, that the two species exchange energy over a distance short compared with the length $L_{T_h} = T_h/|\nabla T_h|$ over which T_h changes. This holds in both layers: λ_{T_h}/L_{T_h} is at most 0.10 at the cathode (at the surface) and 0.021 at the anode. Within about one λ_{T_h} of the surface (0.5 μm at the cathode and 0.1 μm at the anode), the heavy particles are nevertheless not in local equilibrium with the wall, and the boundary condition $T_h(0) = T_{\mathrm{wall}}$ neglects a temperature jump of order $\lambda_{T_h}|\nabla T_h|$, about 0.36 kK at the cathode and 21 K at the anode. Second, the ions must not gain energy from the field faster than they share it with the atoms. At the anode, this condition is satisfied except next to the surface: the effective ion temperature T_{eff} of the two-temperature closure (Appendix B) exceeds T_h by 35 % at the surface, but by more than 10 % only within 0.34 μm of it. In the cathode sheath the condition is not satisfied. The energy $eE\lambda_{T_h}$ that an ion gains from the field between collisions is 8 eV at the surface, against $k_B T_h = 0.3$ eV, and T_{eff} exceeds T_h by a factor of 1.8 at the sheath edge and of 10.6 at the surface; it is more than twice T_h within 0.48 μm of the surface, and more than 10 % above it within 2.1 μm. The random energy of the ions there is not $k_B T_h$. The model retains this heating in the ion mobility, through T_{eff}, but not in the ion pressure, which is evaluated with T_h. Evaluated with T_{eff} instead, the ion pressure gradient would reach 46 % of the electric force in the cathode sheath, instead of 23 %. The ion flux in the inner part of the cathode sheath is therefore uncertain to this extent, in addition to the effect of the neglected inertia. Beyond about 2 μm from the cathode and 0.3 μm from the anode, a single heavy-particle temperature is adequate.

Boltzmann electrons and the Bohm criterion. Simplified sheath models describe the electrons in the sheath by the Boltzmann relation and impose the Bohm criterion on the ions entering it, $u_i \geq u_B$, with $u_B = \sqrt{k_B(T_e + T_i)/M}$. Figure 6(c)–(f) tests both against the computed sheaths. The electron density follows the Boltzmann relation closely. Started at the sheath edge, the relation reproduces n_e to within 1.6 % across the cathode sheath, where n_e falls by a factor of 9,

and to within 1.5 % across the anode sheath, where it falls by a factor of 3.9. Outside the anode sheath, it still holds to within 4.2 % up to 10 μm from the surface, but n_e is 12 % below n_B at 30 μm and 19 % below at 100 μm. At the cathode, it fails progressively in the ionization layer, where the electron drag carries the current: n_e is 9 % below n_B at 10 μm and 53 % below at 100 μm. The ions, in contrast, do not satisfy the Bohm criterion at the sheath edge (figure 6(e), (f)). At the cathode they enter the sheath at $u_i = 1690\ \mathrm{m\,s^{-1}}$, against $u_B = 2160\ \mathrm{m\,s^{-1}}$, i.e. at $0.78\,u_B$. They reach u_B only 0.39 μm from the surface, inside the sheath, where the charge separation is already 3.4 %, and arrive at the surface at $2.4\,u_B$. At the anode, the ions enter the sheath at $0.17\,u_B$ and reach the surface at $0.57\,u_B$, never attaining u_B. The Bohm flux $n_i u_B$ at the sheath edge would therefore overestimate the ion flux entering the sheath by 28 % at the cathode and by a factor of 6 at the anode. The Bohm criterion is derived for a collisionless sheath. In these sheaths the ions collide with the atoms throughout the presheath and the sheath, and, except in the inner part of the cathode sheath, their velocity is set by the balance of the electric force and the drag rather than by the potential drop through which they have fallen. This agrees with the conclusion of Semenov et al. [13] for the anode layer.

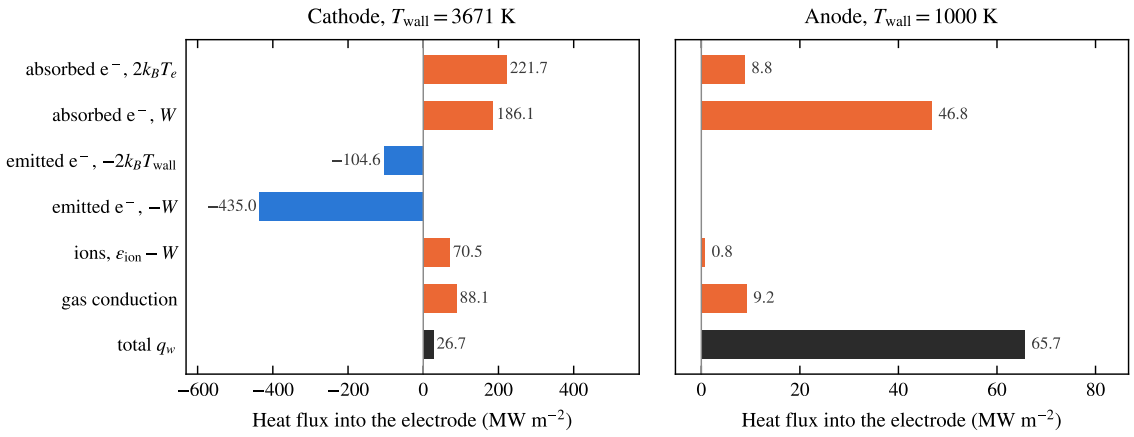


Figure 7. Terms of the heat flux delivered to the electrode, equation (25), positive into the electrode, for the cathode (left) and the anode (right). Orange: heating; blue: emission cooling; black: the net heat flux q_w. At the cathode, q_w equals the conduction $k_w(T_{\mathrm{wall}} - T_0)/L_w$ into the electrode body, equation (24); the anode surface temperature is prescribed.

4.4 Heat flux to the electrode

The terms of the electrode heat flux (25) are shown in figure 7. At the cathode, the heat flux is the small difference of large terms. The absorbed plasma electrons (408 MW m^{-2}), gas conduction (88 MW m^{-2}) and ion recombination (70 MW m^{-2}) heat the surface by 566 MW m^{-2} in total, and emission cools it by 540 MW m^{-2}, so that only $q_w = 26.7$ MW m^{-2} is conducted into the cathode body. The surface temperature, 3670 K, is therefore fixed mainly by the condition that emission cooling nearly balances the heating, and only weakly by the conduction into the cathode body. Emission cooling grows with T_{wall} as $T_{\mathrm{wall}}^2 \exp[-(W - \Delta W)/(k_B T_{\mathrm{wall}})]$, i.e. by about 0.27 % per kelvin at this temperature, so a 5 % change in the heating would shift T_{wall} by only about 20 K, and q_w by less than 1 %. Ion recombination, at $\varepsilon_{\mathrm{ion}} - W = 13.1$ eV per ion, provides 12 % of the heating, although the ions carry only 5.4 % of the current.

At the anode, $q_w = 65.7$ MW m^{-2}, and the absorbed electrons deliver 85 % of it: 46.8 MW m^{-2} as the work function released on entering the metal and 8.8 MW m^{-2} as kinetic energy. Gas conduction contributes 9.2 MW m^{-2} and ion recombination 1.2 %. The anode heat flux is thus set by the electron current, through W, and by the electron temperature at the surface. Since the surface temperature is prescribed, this is the heat that the cooling of the anode has to remove.

Ion impact energy. Equation (25) charges the ions with their recombination energy only, not with the kinetic energy with which they strike the surface. In the model, the ion energy follows from the local field through the two-temperature closure (Appendix B), which presumes that the ions make many collisions over the distance in which the field changes. The cathode sheath does not satisfy this: the ions cross it with only one to two collisions, and their inertia exceeds the drag over its inner part (Section 4.3). The local mean ion energy at the cathode surface is therefore not a reliable impact energy, and its distribution would require a kinetic treatment of the sheath. We therefore give only the maximum impact energy at the cathode: that of an ion that enters the

sheath with the mean energy at the sheath edge, $\frac{3}{2}k_B T_{\text{eff}} + \frac{1}{2}Mv_d^2 = 1.3$ eV (equations (43) and (44)), and crosses it without a collision, gaining $e\phi_s = 3.6$ eV. This maximum is 4.9 eV per ion, and with the ion flux of table 2 it delivers at most 26 MW m^{-2}, comparable to q_w itself. Added to the heating, however, this term would raise T_{wall} by only about 20 K, since most of it would be removed by emission cooling, and q_w by less than 1 %. At the anode, where the ions carry 0.7 % of the current and fall through only 0.60 V in a collision-dominated sheath, their kinetic energy is negligible.

4.5 Dependence on the bulk temperature and the current density

The two cases were repeated with the bulk temperature varied at fixed current density, $T_{\text{bulk}} = 19\,000$–$23\,000$ K for the cathode and 11 000–14 000 K for the anode (figure 8), and with the current density varied at fixed bulk temperature, $J_{\text{app}} = 8 \times 10^7$–$1.5 \times 10^8$ A m^{-2} for the cathode and $|J_{\text{app}}| = 7 \times 10^6$–$2 \times 10^7$ A m^{-2} for the anode (figure 9). Everything else is unchanged. In particular, Λ is recalibrated for each bulk state through equation (15), which gives column radii of 0.33–0.66 mm at the cathode and 1.3–7.3 mm at the anode.

Cathode. The heat flux to the cathode hardly changes: $q_w = 26.3$–27.3 MW m^{-2} in every case (figures 8(a) and 9(a)), while T_{wall} varies only between 3633 and 3726 K. This follows from the heat balance (24). With the electrode parameters used here, a change of q_w by 1 MW m^{-2} requires a change of T_{wall} by 100 K, which would change the emission cooling by about 30 %. The heating of the surface by the absorbed electrons, the ions and gas conduction does depend on the plasma: it increases from 488 to 735 MW m^{-2} between $J_{\text{app}} = 8 \times 10^7$ and 1.5×10^8 A m^{-2}. The surface temperature then rises by only 93 K, and the additional heating is removed by emission cooling. Within these ranges, the heat flux conducted into the cathode is therefore set by the electrode, through W, A_R, k_w, L_w and T_0, rather than by the plasma. The plasma adjusts instead. With increasing current density, the sheath voltage rises from 3.1 to 5.0 V and the ion fraction of the current at the surface from 5.0 to 6.2 % (figure 9). The electron temperature at the surface rises from 16.7 to 21.4 kK, and the voltage across the domain from 11.4 to 15.4 V. The emitted current exceeds the arc current in every case, but by less as the current grows, from 1.77 to 1.44 times the arc current. Above 1.5×10^8 A m^{-2}, no steady state was reached with this cathode heat balance. Increasing the bulk temperature at fixed current acts on the sheath in the opposite direction (figure 8). Between 19 000 and 23 000 K, ϕ_s decreases from 4.1 to 3.2 V, j_i/j from 5.8 to 5.1 % and T_e at the surface from 18.9 to 17.6 kK, while the voltage across the domain, $\phi(L)$, falls from 13.1 to 12.1 V. The sheath width changes by less than 10 % in either sweep (figures 8(g) and 9(g)): it decreases from 0.60 to 0.55 μm as the current density increases, and from 0.59 to 0.57 μm as the bulk temperature increases. The larger sheath voltage at higher current is sustained across a nearly unchanged width by a denser ion space charge: the ion density at the surface rises from 5.5×10^{21} to 9.2×10^{21} m^{-3}.

Anode. The anode heat flux increases almost in proportion to the current density, from 48.8 to 126 MW m^{-2} between $|J_{\text{app}}| = 7 \times 10^6$ and 2×10^7 A m^{-2} (figure 9(b)), i.e. from 7.0 to 6.3 W per ampere. Of this, the work function released by the absorbed electrons contributes $W|J_{\text{app}}|/e$ to within 1 %, i.e. 4.7 W per ampere. Their kinetic energy, $2k_B T_e$ per electron with T_e at the surface between 5.1 and 5.7 kK, contributes another 0.9–1.0 W per ampere. Gas conduction (9.0–11.1 MW m^{-2}) and ion recombination (below 1.3 MW m^{-2}) contribute the rest; at the lower current densities, gas conduction makes up a larger share of q_w, which is why the heat flux per ampere falls with the current. At fixed current, the bulk temperature has a much weaker effect: q_w rises from 60.4 to 65.7 MW m^{-2} between 11 000 and 14 000 K (figure 8(b)). The electron terms are fixed by the current and by T_e at the surface, which stays between 5.0 and 5.3 kK, and the increase comes mostly from gas conduction, which rises from 4.4 to 9.2 MW m^{-2}. In every anode case, the ions carry less than 1 % of the current. The anode lies below the bulk-plasma boundary by $\phi(L) = 1.4$ to 3.4 V as T_{bulk} increases from 11 000 to 14 000 K. With increasing current density this drop decreases, from 4.0 V at 7×10^6 to 2.2 V at 2×10^7 A m^{-2}. The potential at the edge of the thin space-charge layer stays between 0.23 and 0.75 V. The width of this layer decreases with the electron density at the surface. It decreases from 0.90 to 0.63 μm as the current density increases from 7×10^6 to 2×10^7 A m^{-2} (figure 9(h)), while n_e at the surface rises from 4.0×10^{20} to 1.1×10^{21} m^{-3}: somewhat more slowly than $n_e^{-1/2}$, the scaling of the Debye length. At fixed current, n_e at the surface changes by at most 4 % with the bulk temperature, and the width stays between 0.72 and 0.81 μm (figure 8(h)).

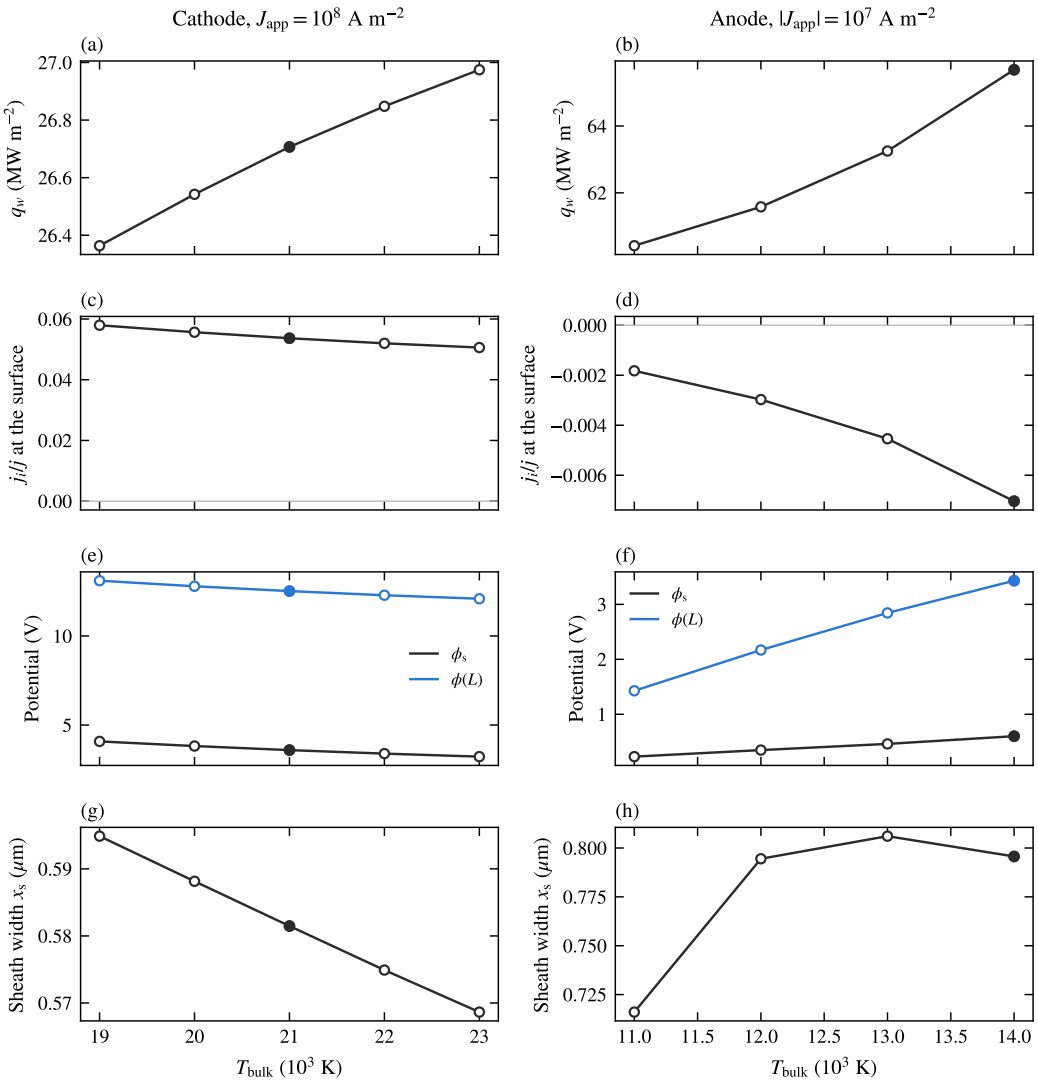


Figure 8. Electrode quantities against the bulk temperature T_{bulk}, for the cathode at $J_{\mathrm{app}} = 10^8$ A m^{-2} (left) and the anode at $|J_{\mathrm{app}}| = 10^7$ A m^{-2} (right). Filled symbols: the cases of table 2. (a), (b) Heat flux into the electrode q_w, equation (25). (c), (d) Ion fraction of the current density at the surface, positive for ions flowing into the cathode; at the anode the ions also flow into the surface, against the current. (e), (f) Potential at the sheath edge ϕ_{s} and at the bulk-plasma boundary $\phi(L)$, relative to the electrode. (g), (h) Sheath width x_{s}, the distance from the surface at which $|n_i - n_e|/n_i$ falls to 1 %.

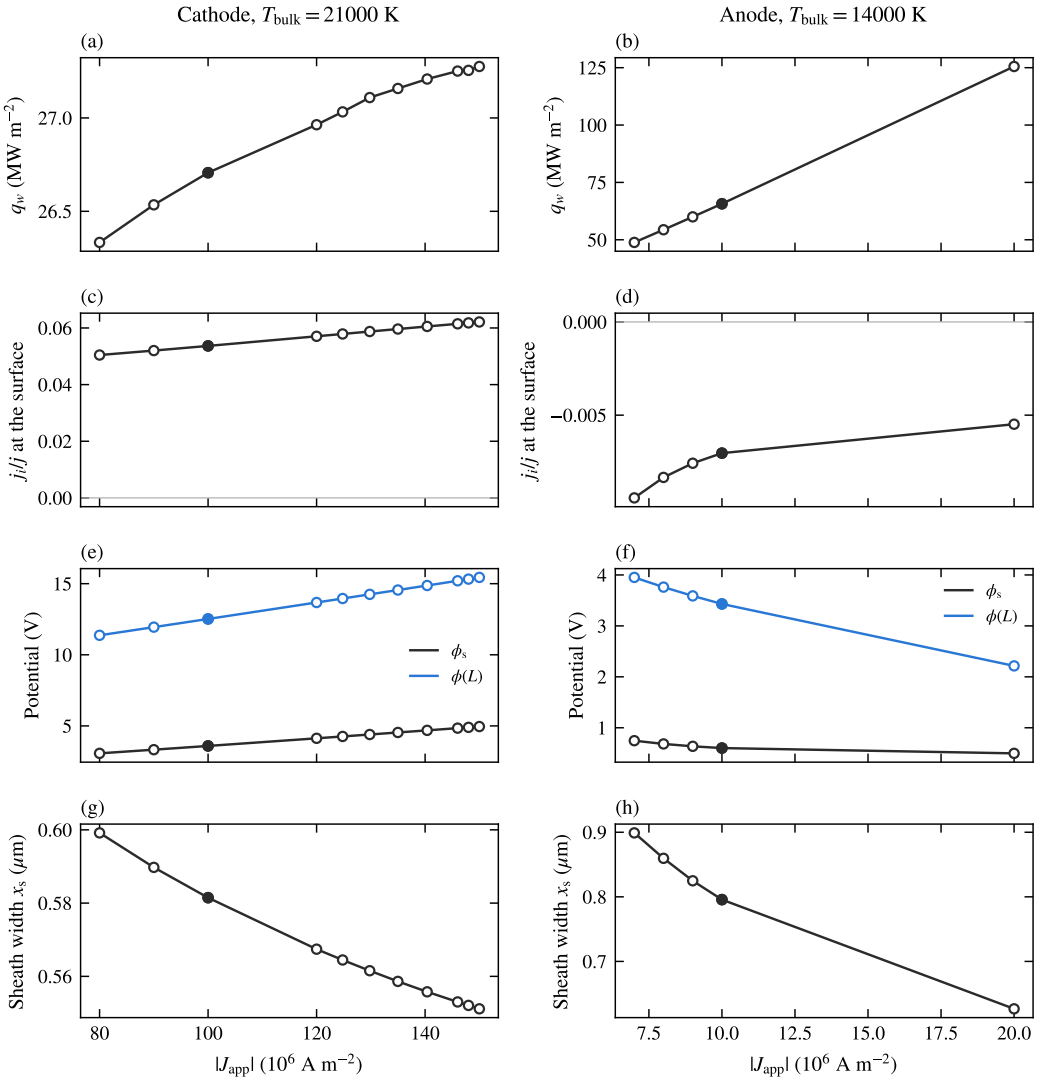


Figure 9. As figure 8, against the current density $|J_{\rm app}|$, for the cathode at $T_{\rm bulk} = 21\,000$ K (left) and the anode at $T_{\rm bulk} = 14\,000$ K (right).

4.6 Relaxation times

To estimate how quickly the near-electrode layers follow a change of the arc, each steady state of Sections 4.1–4.5 was perturbed by a step of +2 % in $|J_{\mathrm{app}}|$, and the time-dependent equations were integrated to the new steady state. The surface temperature and the radial-loss coefficient Λ were held at their steady values, since the electrode body responds on a time scale of seconds and Λ represents the column outside the domain. The response of the ion current density j_i and of the heat flux q_w at the surface is characterized by the time to steady state t_{99}, after which the quantity stays within 1 % of its total change.

In a one-dimensional domain, the step also excites an axial acoustic oscillation of the heavy gas between the electrode and the boundary $x = L$. In a three-dimensional arc, such a pressure disturbance is relieved radially within $R/c \approx 0.1$ μs. The axial acoustics are therefore removed by multiplying the inertia of the heavy gas, the left-hand side of equation (18), by $\epsilon = 10^{-4}$. This raises the sound speed to $c/\sqrt{\epsilon}$ and leaves the steady state, with $\mathbf{u} = 0$, unchanged; t_{99} varies by less than 3 % for $10^{-6} \le \epsilon \le 10^{-3}$. At the anode, the acoustics do not affect the response, and the physical sound speed is retained.

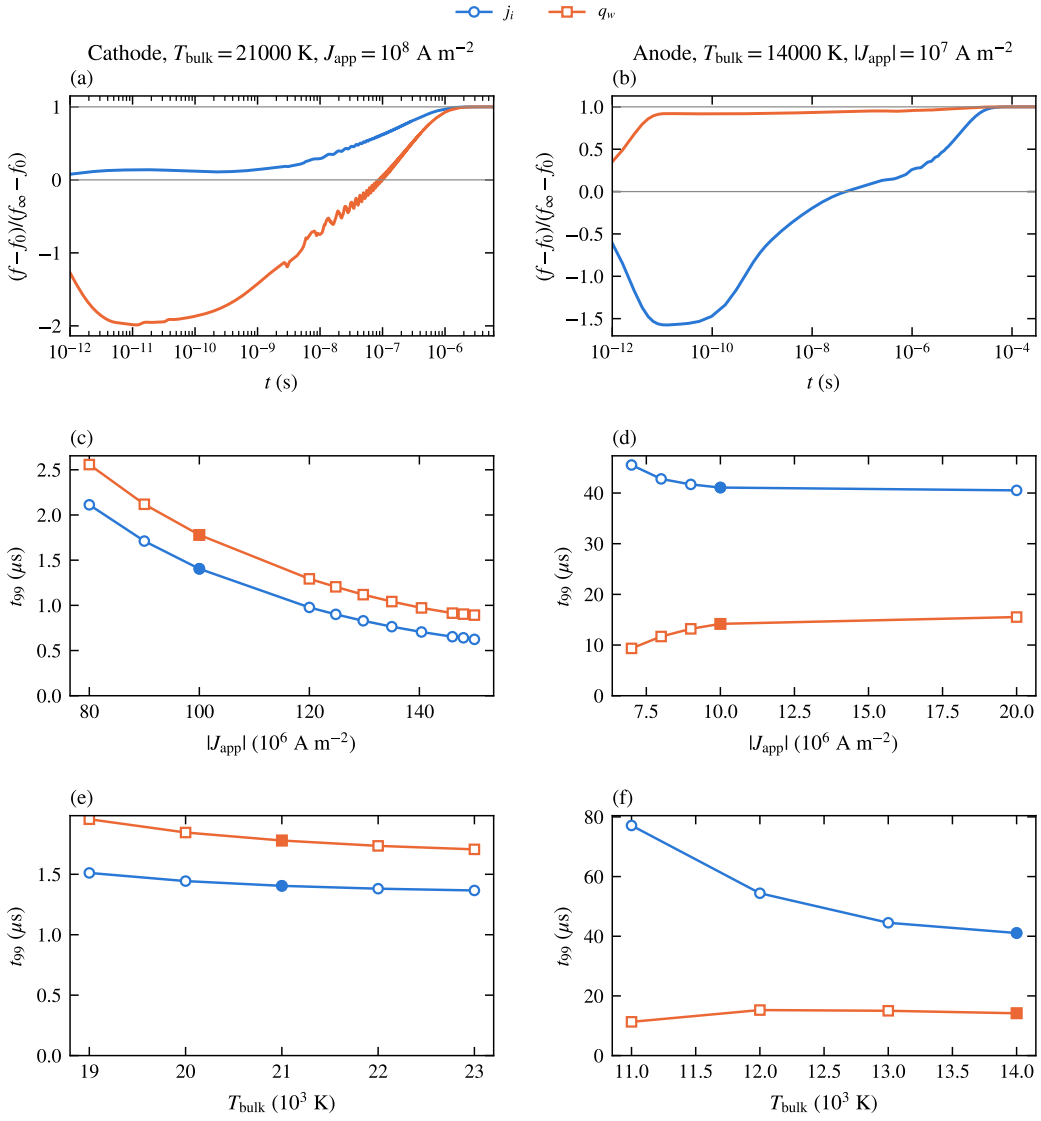


Figure 10. Response of the electrode quantities to a +2 % step of $|J_{\mathrm{app}}|$: cathode (left, axial acoustics removed) and anode (right). Blue: ion current density j_i at the surface; orange: heat flux q_w into the surface. (a), (b) Normalized response for the conditions of table 2, f_0 and f_∞ being the initial and the final values. (c), (d) Time to steady state t_{99} against $|J_{\mathrm{app}}|$ at the T_{bulk} of table 2. (e), (f) t_{99} against T_{bulk} at the $|J_{\mathrm{app}}|$ of table 2. Filled symbols: the conditions of table 2.

Cathode. The response has two stages (figure 10(a)). About 14 % of the change of j_i occurs within 10 ps; the rest follows within about 1.5 μs, $t_{99} = 1.40$ μs, the final approach being

exponential with a relaxation time of 0.39 μs. The heat flux first moves in the opposite direction and settles later, $t_{99} = 1.78$ μs. The cathode layer responds faster at higher current density: between 8×10^7 and 1.5×10^8 A m^{-2}, t_{99} falls from 2.1 to 0.62 μs for j_i and from 2.6 to 0.89 μs for q_w (figure 10(c)). The bulk temperature changes t_{99} by less than 15 % between 19 000 and 23 000 K (figure 10(e)).

Anode. The anode layer relaxes about 24 times more slowly, with a relaxation time of 9.3 μs, and its two quantities behave differently (figure 10(b)). Because the dominant electron terms of the heat flux (25) are carried by the current (Section 4.4), 92 % of the change of q_w occurs within 1 ns, and $t_{99} = 14$ μs. The ion current density first moves in the opposite direction and needs $t_{99} = 41$ μs. For j_i, t_{99} decreases from 77 to 41 μs as T_{bulk} rises from 11 000 to 14 000 K, and only from 46 to 41 μs over the current range; that of q_w stays between 9 and 16 μs in all cases (figure 10(d), (f)).

The cathode layer can therefore be treated as quasi-steady for changes of the arc slower than a few microseconds. At the anode, the heat flux follows the current within about 15 μs, but the ion current needs 40–80 μs.

5 Conclusion

A unified drift-diffusion–Poisson model of the near-electrode layers of an atmospheric-pressure argon arc has been presented. A single set of equations is solved from the electrode surface, through the space-charge sheath and the ionization layer, to the bulk plasma: the continuity equations with fluxes derived from the species momentum balances, including thermal diffusion and the collisional drag between species, separate electron and heavy-particle energy equations, a heavy-gas momentum equation and Poisson's equation. Adaptive mesh refinement resolves the sheath on a fraction of the Debye length. The surface temperature of the cathode follows from a heat balance that includes thermionic emission with Schottky lowering. The model reproduces the near-cathode layer computed by Almeida et al. [11] and the near-anode layer computed by Semenov et al. [13]: the temperatures agree within 3–5 %, the charged-particle densities within 15 % outside the cathode sheath, and the ion fraction of the current at the surface within 5 %.

For a cathode at $T_{\mathrm{bulk}} = 21\,000$ K and 10^8 A m^{-2}, and an anode at 14 000 K and 10^7 A m^{-2}, with the radial heat loss of the column calibrated on the bulk state, the main results are as follows.

- The two layers have opposite structures. The cathode has an ion-rich sheath 0.58 μm wide with a voltage of 3.6 V, followed by an ionization layer of about 10^{-4} m in which the gas is heated, the plasma becomes almost fully ionized and the charged-particle density, driven by the hot electrons, approaches its equilibrium value at the heavy-particle temperature only at the end of the layer. Joule heating is the dominant electron energy source throughout. The anode has practically no sheath (0.60 V across 0.80 μm). Its 3.4 V retarding drop is spread over the quasi-neutral layer, the charged particles are supplied by diffusion at densities far above ionization equilibrium, and $T_e - T_h$ reaches 4.1 kK at the surface. The electrons lose energy to the retarding field, and this loss is supplied by the electron energy flux from the column, mainly through conduction and the thermal-diffusion heat flux close to the anode.

- The electrons cross both sheaths with fewer than 0.2 collisions, yet their density follows the Boltzmann relation within about 2 %, so that the electron force balance remains adequate there. The ions cross 1.8 (cathode) and 7.7 (anode) mean free paths in the sheaths and do not satisfy the Bohm criterion at the sheath edge, where they move at 0.78 and 0.17 times the Bohm velocity. A Bohm flux at the sheath edge would overestimate the ion flux by 28 % at the cathode and by a factor of 6 at the anode.

- The cathode heat flux, 27 MW m^{-2}, is the small difference between the heating of the surface (566 MW m^{-2}) and emission cooling (540 MW m^{-2}). Because emission cooling depends steeply on T_{wall}, the heat flux into the cathode body is set by the electrode rather than by the plasma: it changed by less than 4 % over all the cathode cases, while the heating of the surface changed by 50 %. The anode heat flux, 66 MW m^{-2}, is 85 % electronic and increases nearly in proportion to the current, at 6.3–7.0 W per ampere, of which 4.7 W per ampere is the work function ($W = 4.65$ eV, copper).

- Because the ions cross the cathode sheath with only one to two collisions, a local estimate of their impact energy is not reliable, and only its maximum, for an ion crossing the sheath without a collision, has been given: 4.9 eV, corresponding to at most 26 MW m^{-2}. Owing to the stiffness of emission cooling, this term would change the cathode surface temperature by only about 20 K.

- Over the ranges of bulk temperature and current density studied, the cathode sheath width stays between 0.55 and 0.60 μm, its larger voltage at higher current being sustained by a denser ion space charge. The anode sheath width decreases with the electron density at the surface, from 0.90 to 0.63 μm as the current density increases from 7×10^6 to 2×10^7 A m^{-2}.

- After a change of the current, the cathode layer reaches a new steady state within 0.6–2.6 μs. At the anode, the heat flux follows within about 15 μs, but the ion current needs 40–80 μs.

The local, inertialess transport description is marginal within about 0.3 μm of the cathode, where the inertia of the ions exceeds their drag. In the same region, the field heats the ions well above the atom temperature, so that the single heavy-particle temperature does not describe their random energy, and the ion energy distribution at the cathode requires a kinetic treatment of the sheath. Elsewhere, ions and atoms exchange energy over distances short compared with the gradients of T_h, and a single heavy-particle temperature is adequate. The present one-dimensional calculations represent the radial losses of the column through a calibrated sink. Since the layers settle within microseconds at the cathode and tens of microseconds at the anode, and are

characterized by the bulk state and the current density, the model can supply electrode boundary conditions, such as the voltage drop, the current partition and the heat flux to the electrode, to LTE models of the arc column.

A Quantities used in the scale estimates

The scales of table 1 are evaluated in LTE at atmospheric pressure, $T_e = T_h = T$, with the composition given by the Saha equation (internal partition functions summed over the low-lying levels of Ar and Ar^+), quasi-neutrality and Dalton's law. Following Almeida et al. [11], the averaged electron–atom and electron–ion momentum-transfer cross sections are

$$\overline{Q}_{en} = \frac{1}{2}\int_0^\infty x^2 e^{-x}\, Q^{(1)}(x\, k_B T_e)\, dx, \qquad \overline{Q}_{ei} = \frac{e^4 \ln\Lambda}{32\pi\varepsilon_0^2 (k_B T_e)^2}, \tag{29}$$

where $Q^{(1)}$ is the argon momentum-transfer cross section of the LXCat database [44], $x = \varepsilon/(k_B T_e)$ is the electron energy ε reduced by $k_B T_e$, and $\ln\Lambda$ is the Coulomb logarithm, equation (39). The ion–ion cross section $\overline{Q}_{ii}$ is $\overline{Q}_{ei}$ evaluated at T_h, and the ion–atom cross section $\overline{Q}_{in}$ is given by equation (41). The ambipolar diffusion coefficient is

$$D_a = \left(1 + \frac{T_e}{T_h}\right) D_{in}, \qquad D_{in} = \frac{3\pi}{32}\frac{\bar{C}_{in}}{(n_i + n_n)\,\overline{Q}_{in}}, \qquad \bar{C}_{in} = \sqrt{\frac{8 k_B T_h}{\pi\, m_{in}}}, \tag{30}$$

where D_{in} is the ion–atom diffusion coefficient [11] and $m_{in} = M/2$ is the ion–atom reduced mass. The reaction time τ_{rxn} uses the rate coefficients of Almeida et al. [11], from Benilov and Naidis [38].

B Transport coefficients

This appendix describes how the transport and energy-exchange coefficients that close the model of Section 2.2 are calculated, and compares the resulting conductivities with the transport properties of argon in LTE.

B.1 Electrons

Following Mitchner and Kruger [45], the electron current density $\mathbf{J}_e = -e\boldsymbol{\Gamma}_e$ and the electron heat flux $\mathbf{q}_e$ are written as

$$\mathbf{J}_e = \sigma\left(\mathbf{E} + \nabla p_e/(e n_e)\right) + \varphi\,\nabla T_e, \tag{31}$$

$$\mathbf{q}_e = -\lambda \nabla T_e + \psi\,\mathbf{J}_e. \tag{32}$$

Here σ is the electrical conductivity, φ is the thermal-diffusion coefficient, λ is the ordinary (zero-current) thermal conductivity, and ψ is the energy transported by the current per unit charge. Comparing equations (31) and (32) with the electron momentum balance (3) and the energy flux (13) gives

$$\bar{\nu}_e^M = \frac{n_e e^2}{m_e \sigma}, \qquad C_e = \frac{e}{k_B}\frac{\varphi}{\sigma}, \qquad \kappa_e = \lambda, \qquad -\frac{e\psi}{k_B T_e} = \frac{5}{2} + C_e. \tag{33}$$

Mitchner and Kruger [45] give σ, φ, and λ in the weakly and fully ionized limits. Substituting them into equation (33) gives the collision frequencies and thermal-diffusion coefficients in each limit.

Weakly ionized limit. Electron–neutral collisions dominate, and the electrons behave as a Lorentz gas whose distribution function has a Maxwellian isotropic part $f_M(T_e)$. Let C be the electron speed, $\varepsilon = \frac{1}{2} m_e C^2$ the electron energy, $x = \varepsilon/(k_B T_e)$ the reduced energy, and $\nu_{en} = n_n C Q^{(1)}(\varepsilon)$ the energy-dependent momentum-transfer frequency, where $Q^{(1)}$ is the momentum-transfer cross section. A two-term solution of the electron Boltzmann equation then gives

$$\bar{\nu}_{en}^M = \frac{\langle C^2\rangle_0}{\langle C^2/\nu_{en}\rangle_0}, \qquad C_{en} = \langle x\rangle_\nu - \frac{5}{2}, \qquad \lambda_{en} = \frac{n_e k_B^2 T_e}{m_e \bar{\nu}_{en}^M}\left[\langle x^2\rangle_\nu - \langle x\rangle_\nu^2\right], \tag{34}$$

where $\bar{\nu}_{en}^M$ is the momentum-transfer average of ν_{en}, $\langle\cdot\rangle_0$ is the Maxwellian average, and $\langle\cdot\rangle_\nu$ is the transport average with the weight $C^4 f_M/\nu_{en}$:

$$\langle g\rangle_0 = 4\pi\int_0^\infty g\, C^2 f_M\, dC, \qquad \langle g\rangle_\nu = \frac{\langle g\, C^2/\nu_{en}\rangle_0}{\langle C^2/\nu_{en}\rangle_0}. \tag{35}$$

The elastic energy exchange in equations (12) and (16) uses instead the energy-exchange average

$$\bar{\nu}_{en}^{E} = \frac{\langle C^2 \nu_{en} \rangle_0}{\langle C^2 \rangle_0}. \tag{36}$$

All averages are computed from the argon momentum-transfer cross section $Q^{(1)}$ of the LXCat database [44].

Fully ionized limit. For singly charged ions, the Spitzer–Härm solution gives [45]

$$\bar{\nu}_{ei}^{M} = \frac{\bar{\nu}_{ei}^{E}}{1.975}, \qquad C_{ei} = \frac{1.389}{1.975} = 0.703, \qquad \lambda_{ei} = 3.203\,\frac{n_e k_B^2 T_e}{m_e \bar{\nu}_{ei}^{E}}, \tag{37}$$

where $\bar{\nu}_{ei}^{E}$ is the energy-exchange average (36) of the Coulomb collision frequency [45],

$$\bar{\nu}_{ei}^{E} = n_i\,\frac{4\sqrt{2\pi}}{3}\left(\frac{m_e}{k_B T_e}\right)^{3/2}\left(\frac{e^2}{4\pi\varepsilon_0 m_e}\right)^{2}\ln\Lambda, \tag{38}$$

and $\ln\Lambda$ is the Coulomb logarithm,

$$\ln\Lambda = \max\left(2,\ \ln\frac{\lambda_{De}}{b_{90}}\right), \qquad \lambda_{De} = \sqrt{\frac{\varepsilon_0 k_B T_e}{n_e e^2}}, \qquad b_{90} = \frac{e^2}{12\pi\varepsilon_0 k_B T_e}. \tag{39}$$

Here λ_{De} is the electron Debye length and b_{90} is the 90°-deflection impact parameter. The floor at 2 keeps $\ln\Lambda$, and hence $\bar{\nu}_{ei}^{E}$, positive when the plasma is not weakly coupled.

At intermediate ionization, the two limits are combined. The momentum-transfer frequencies are added, the thermal conductivities are combined in series (their thermal resistances add), and the thermal-diffusion coefficient is the mean of the two limits weighted by the energy-exchange frequencies:

$$\bar{\nu}_{e}^{M} = \bar{\nu}_{en}^{M} + \bar{\nu}_{ei}^{M}, \qquad \frac{1}{\kappa_e} = \frac{1}{\lambda_{en}} + \frac{1}{\lambda_{ei}}, \qquad C_e = \frac{C_{en}\bar{\nu}_{en}^{E} + C_{ei}\bar{\nu}_{ei}^{E}}{\bar{\nu}_{en}^{E} + \bar{\nu}_{ei}^{E}}. \tag{40}$$

B.2 Ions

The ion–neutral transport is governed by resonant charge exchange. The momentum-transfer cross section is taken from the fit [38]

$$\overline{Q}_{in} = 2\left[7\times10^{-10} - 0.6\times10^{-10}\ln\left(\frac{2k_B T_h}{e}\right)\right]^{2}\ \mathrm{m}^2, \tag{41}$$

with $2k_B T_h/e$ in eV. The corresponding low-field mobility is obtained from the Mason–Schamp equation [46],

$$\mu_i^0 = \frac{1.104\times10^{5}}{\overline{Q}_{in}\sqrt{T_h}\,n_n}\ \ \mathrm{m^2\,V^{-1}\,s^{-1}} \qquad (\overline{Q}_{in}\text{ in m}^2,\ T_h\text{ in K},\ n_n\text{ in m}^{-3}). \tag{42}$$

To account for the effect of strong electric fields on the ion mobility, a closed-form two-temperature Wannier formulation [47, 48] is used, in which the drift speed v_d and the effective ion temperature T_{eff} are determined self-consistently:

$$v_d^2 = \frac{\sqrt{1+4\mathcal{A}\mathcal{B}}-1}{2\mathcal{B}}, \qquad \mathcal{A} = \left(\mu_i^0|\mathbf{E}|\right)^2, \qquad \mathcal{B} = \frac{M}{3k_B T_h}, \tag{43}$$

$$T_{\mathrm{eff}} = T_h + \frac{M v_d^2}{3k_B}, \qquad \mu_{in} = \mu_i^0\sqrt{\frac{T_h}{T_{\mathrm{eff}}}}. \tag{44}$$

The ion–neutral momentum-transfer frequency is then obtained from the mobility as

$$\bar{\nu}_{in}^{M} = \frac{e}{M\mu_{in}}. \tag{45}$$

Figure 11 compares the reduced mobility $K_0 = \mu n_n/N_0$ (with $N_0 = 2.687\times10^{25}$ m^{-3}) of the single-temperature mobility μ_i^0, equation (42), and the two-temperature mobility μ_{in}, equation (44), with the Monte Carlo results of Nikitović and Raspopović [49] for Ar^+ in Ar at 300 K. These Monte Carlo results agree with the measured drift-tube mobilities [49]. At T_h fixed, K_0 depends only on E/N. At low field, both formulations give $K_0 = 1.54$ cm^2 V^{-1} s^{-1}, 2 % below

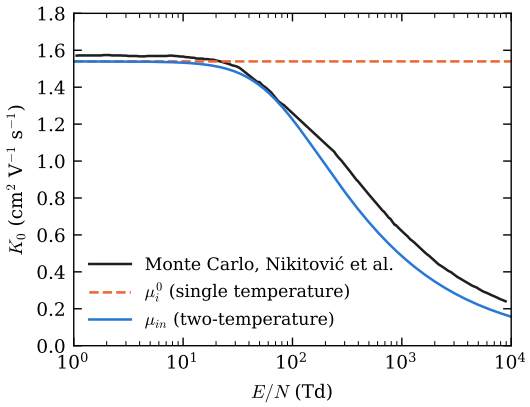


Figure 11. Reduced mobility of Ar^+ in Ar at 300 K as a function of the reduced field E/N. The single-temperature mobility μ_i^0, equation (42) (dashed), and the two-temperature mobility μ_{in}, equation (44) (solid blue), are compared with the Monte Carlo results of Nikitović and Raspopović [49] (black), digitized from their figure 7.

the Monte Carlo value. Above about 30 Td, the single-temperature mobility stays constant and overestimates K_0 by a factor of 1.6 at 300 Td and 4 at 3000 Td. The two-temperature mobility reproduces the decrease of K_0 with E/N, and lies 15 % below the Monte Carlo value at 300 Td and 26 % below at 3000 Td. Most of this remaining difference comes from evaluating $\overline{Q}_{in}$ at T_h instead of T_{eff}. Evaluating it at T_{eff} brings the two-temperature mobility within 7 % of the Monte Carlo values from 1 to 7000 Td, but then T_{eff} has to be found iteratively instead of in closed form.

The ion–electron frequency follows from momentum conservation, equation (6),

$$\bar{\nu}_{ie}^M = \frac{m_e n_e}{M n_i}\,\bar{\nu}_{ei}^M, \tag{46}$$

with $\bar{\nu}_{ei}^M$ given by equation (37). The ions take up the electron–ion part of the electron thermal force, so their thermal coefficient is the C_{ei} term of equation (40),

$$C_i = \frac{C_{ei}\bar{\nu}_{ei}^E}{\bar{\nu}_{en}^E + \bar{\nu}_{ei}^E}. \tag{47}$$

The ion contribution κ_i to the heavy-particle thermal conductivity is calculated as in Almeida et al. [11].

B.3 Heavy species

The atom frequencies follow from momentum conservation, equation (6),

$$\bar{\nu}_{ni}^M = \frac{n_i}{n_n}\,\bar{\nu}_{in}^M, \qquad \bar{\nu}_{ne}^M = \frac{m_e n_e}{M n_n}\,\bar{\nu}_{en}^M. \tag{48}$$

The atoms take up the electron–neutral part of the electron thermal force, so their thermal coefficient is the C_{en} term of equation (40),

$$C_n = \frac{C_{en}\bar{\nu}_{en}^E}{\bar{\nu}_{en}^E + \bar{\nu}_{ei}^E}, \tag{49}$$

and equations (47) and (49) satisfy $C_i + C_n = C_e$, as required by equation (6). The heavy-particle thermal conductivity $\kappa_h = \kappa_n + \kappa_i$, the sum of the atom and ion contributions, is calculated as in Almeida et al. [11].

B.4 Comparison with LTE transport properties

The electrical conductivity $\sigma = n_e e^2/(m_e \bar{\nu}_e^M)$ and the thermal conductivity obtained with these coefficients are compared with the transport properties of argon in local thermodynamic equilibrium (LTE) in figure 12. The coefficients are evaluated with $T_e = T_h = T$ and the Saha composition at atmospheric pressure, and compared with the data tabulated by Boulos et al. [1]. Along the axis, the model has no reactive thermal conductivity: the energy carried by the diffusion of ion pairs and released on recombination appears through the ambipolar flux and the ionization

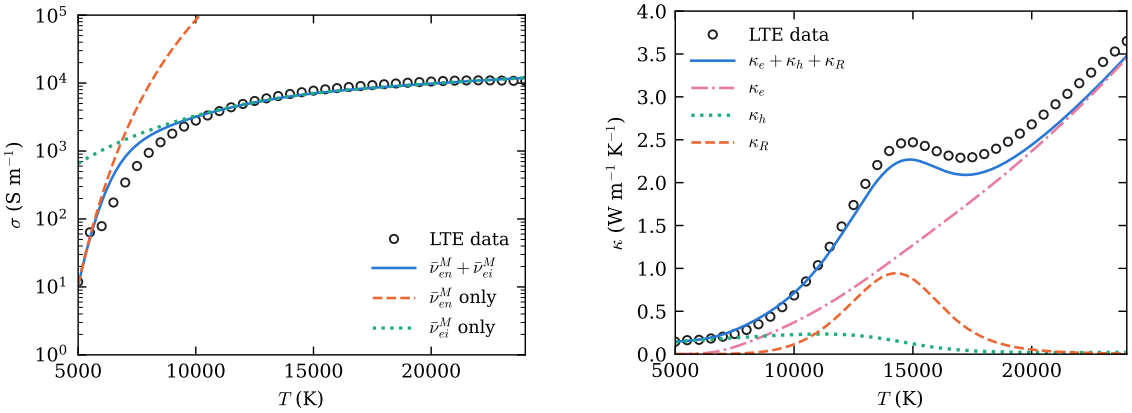


Figure 12. Transport coefficients of argon in LTE at atmospheric pressure, computed with the rules of equation (40) (lines), compared with the LTE data of Boulos et al. [1] (symbols). Left: electrical conductivity $\sigma = n_e e^2/(m_e \bar{\nu}_e^M)$, with the electron–neutral and electron–ion limits shown separately. Right: thermal conductivity, the sum of the electron conductivity κ_e, the heavy-particle conductivity κ_h [11], and the reactive conductivity κ_R computed with Mutation++ [50].

source (κ_R enters only the radial loss (14)). For the comparison, the reactive conductivity κ_R computed with Mutation++ [50] is therefore added to $\kappa_e + \kappa_h$.

The electrical conductivity agrees well with the LTE data over most of the temperature range, with only minor deviations in the low-temperature regime where electron–neutral collisions dominate. At intermediate temperatures, however, the additive rule for the collision frequencies overestimates the conductivity. This discrepancy arises because, in argon, the slow electrons associated with the Ramsauer minimum contribute substantially to the current in the electron–neutral limit, while these same electrons are strongly scattered by ions. Simply adding the averaged electron–neutral and electron–ion collision frequencies therefore fails to capture the correlation between the two collision processes. The thermal conductivity also shows good agreement with the reference data in the high-temperature regime, with a modest overprediction at intermediate temperatures.

Acknowledgments
To be completed.

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